Discrete Mathematics

- **Chapter 1. The Foundations: Logic and Proofs**
- Chapter 2. Basic Structures: Sets, Functions, Sequences, Sums, and Matrices
- Chapter 6. Counting
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The Foundations: Logic and Proofs

Chapter 1

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1.7 Introduction to Proofs

1.8 Proof Methods and Strategy

- Negation $\neg$
- Conjunction $\wedge$
- Disjunction $\vee$
- Implication $\rightarrow$
- Biconditional $\leftrightarrow$

p	$\neg p$
T	F
F	T

p	q	$P \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

p	q	$P \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

p	q	$P \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F

p	q	$P \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Different Ways of Expressing $p \rightarrow q$

if p**, then** q

if p, q

p **implies** q

p **only if** q

q **when** p

q **unless** $\neg p$

q **if** p

q **whenever** p

q **follows from** p

p **is sufficient for** q

q **is necessary for** p

a necessary condition for p **is** q

a sufficient condition for q **is** p

TABLE 6 Logical Equivalences.

Equivalence	*Name*
$p \wedge \mathbf{T} \equiv p$ $p \vee \mathbf{F} \equiv p$	Identity laws
$p \vee \mathbf{T} \equiv \mathbf{T}$ $p \wedge \mathbf{F} \equiv \mathbf{F}$	Domination laws
$p \vee p \equiv p$ $p \wedge p \equiv p$	Idempotent laws
$\neg(\neg p) \equiv p$	Double negation law
$p \vee q \equiv q \vee p$ $p \wedge q \equiv q \wedge p$	Commutative laws
$(p \vee q) \vee r \equiv p \vee (q \vee r)$ $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	Associative laws
$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	Distributive laws
$\neg(p \wedge q) \equiv \neg p \vee \neg q$ $\neg(p \vee q) \equiv \neg p \wedge \neg q$	De Morgan's laws
$p \vee (p \wedge q) \equiv p$ $p \wedge (p \vee q) \equiv p$	Absorption laws
$p \vee \neg p \equiv \mathbf{T}$ $p \wedge \neg p \equiv \mathbf{F}$	Negation laws

TABLE 1 Rules of Inference.

Rule of Inference	*Tautology*	*Name*
p $p \to q$ $\therefore q$	$(p \wedge (p \to q)) \to q$	Modus ponens
$\neg q$ $p \to q$ $\therefore \neg p$	$(\neg q \wedge (p \to q)) \to \neg p$	Modus tollens
$p \to q$ $q \to r$ $\therefore p \to r$	$((p \to q) \wedge (q \to r)) \to (p \to r)$	Hypothetical syllogism
$p \vee q$ $\neg p$ $\therefore q$	$((p \vee q) \wedge \neg p) \to q$	Disjunctive syllogism
p $\therefore p \vee q$	$p \to (p \vee q)$	Addition
$p \wedge q$ $\therefore p$	$(p \wedge q) \to p$	Simplification
p q $\therefore p \wedge q$	$((p) \wedge (q)) \to (p \wedge q)$	Conjunction
$p \vee q$ $\neg p \vee r$ $\therefore q \vee r$	$((p \vee q) \wedge (\neg p \vee r)) \to (q \vee r)$	Resolution

De Morgan's Laws for Quantifiers

The reasoning in the table shows that:

$$\neg\forall x P(x) \equiv \exists x \neg P(x)$$

$$\neg\exists x P(x) \equiv \forall x \neg P(x)$$

The rules for negations for quantifiers are called De Morgan's laws for quantifiers.

Negation	***Equivalent Statement***	***When Is Negation True?***	***When False?***
$\neg\exists x P(x)$	$\forall x \neg P(x)$	For every *x, P*(*x*) is false.	There is *x* for which *P*(*x*) is true.
$\neg\forall x P(x)$	$\exists x \neg P(x)$	There is an *x* for which *P* (*x*) is false.	*P*(*x*) is true for every *x*.

1.2.2 Translating English Sentences

Statements in mathematics and the sciences and in natural language often are imprecise or ambiguous.

Steps to convert an English sentence to a statement in propositional logic

- Identify atomic propositions and represent using propositional variables.
- Determine appropriate logical connectives

Translating English Sentences

Example: " If I go to Harry's house or to the country, I will not go shopping."

One Solution:

- *p*: I go to Harry's house.
- *q*: I go to the country.
- *r*: I will go shopping.

If *p* or *q* then not *r*. $\left(p \vee q\right) \rightarrow \neg r$

1.3.1 Introduction

A *tautology* is a proposition which is always true.

- Example: $p \vee \neg p$

A *contradiction* is a proposition which is always false.

- Example: $p \wedge \neg p$

A *contingency* is a proposition which is neither a tautology nor a contradiction, such as p

P	$\neg p$	$p \vee \neg p$	$p \wedge \neg p$
T	F	T	F
F	T	T	F

De Morgan's Laws

Augustus De Morgan 1806-1871

$$\neg(p \wedge q) \equiv \neg p \vee \neg q$$ (the first De Morgan law)

$$\neg(p \vee q) \equiv \neg p \wedge \neg q$$ (the second De Morgan law)

This truth table shows that De Morgan's Second Law holds.

p	q	$\neg p$	$\neg q$	$(p \vee q)$	$\neg(p \vee q)$	$\neg p \wedge \neg q$
T	T	F	F	T	F	F
T	F	F	T	T	F	F
F	T	T	F	T	F	F
F	F	T	T	F	T	T

TABLE 6 Logical Equivalences.

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$(p \vee q) \vee r \equiv p \vee (q \vee r)$ $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	Associative laws
$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	Distributive laws
$\neg(p \wedge q) \equiv \neg p \vee \neg q$ $\neg(p \vee q) \equiv \neg p \wedge \neg q$	De Morgan's laws
$p \vee (p \wedge q) \equiv p$ $p \wedge (p \vee q) \equiv p$	Absorption laws
$p \vee \neg p \equiv \mathbf{T}$ $p \wedge \neg p \equiv \mathbf{F}$	Negation laws

More Logical Equivalences

TABLE 7 Logical Equivalences Involving Conditional Statements.

$$p \to q \equiv \neg p \lor q$$
$$p \to q \equiv \neg q \to \neg p$$
$$p \lor q \equiv \neg p \to q$$
$$p \land q \equiv \neg(p \to \neg q)$$
$$\neg(p \to q) \equiv p \land \neg q$$
$$(p \to q) \land (p \to r) \equiv p \to (q \land r)$$
$$(p \to r) \land (q \to r) \equiv (p \lor q) \to r$$
$$(p \to q) \lor (p \to r) \equiv p \to (q \lor r)$$
$$(p \to r) \lor (q \to r) \equiv (p \land q) \to r$$

TABLE 8 Logical Equivalences Involving Biconditional Statements.

$$p \leftrightarrow q \equiv (p \to q) \land (q \to p)$$
$$p \leftrightarrow q \equiv \neg p \leftrightarrow \neg q$$
$$p \leftrightarrow q \equiv (p \land q) \lor (\neg p \land \neg q)$$
$$\neg(p \leftrightarrow q) \equiv p \leftrightarrow \neg q$$

1.5.4 Translating Mathematical Statements into Nested Quantifiers

Example : Translate "The sum of two positive integers is always positive" into a logical expression.

Solution:

1. Rewrite the statement to make the implied quantifiers and domains explicit:

   "For every two integers, if these integers are both positive, then the sum of these integers is positive."

2. Introduce the variables *x* and *y*, and specify the domain, to obtain:

   "For all positive integers *x* and *y*, *x* + *y* is positive."

3. The result is: $\forall x \forall y \big((x>0)\wedge(y>0)\big)\rightarrow(x+y>0)$

   where the domain for both variables consists of all integers.

Proofs

Direct Proofs

Indirect Proofs

- Proof by Contraposition
- Proof by Contradiction

1.8.2 Exhaustive Proof and Proof by Cases

To prove a conditional statement of the form:

$$(p_1 \vee p_2 \vee \ldots \vee p_n) \to q$$

Use the tautology

$$[(p_1 \vee p_2 \vee \ldots \vee p_n) \to q] \leftrightarrow$$
$$[(p_1 \to q) \wedge (p_2 \to q) \wedge \ldots \wedge (p_n \to q)]$$

Each of the implications $p_i \to q$ is a *case*.

An exhaustive proof is a special type of proof by cases where each case involves checking a single example.

Extra exercises

Using the definitions of even integer and odd integer, give a direct proof that this statement is true for all integers n: If n is odd, then 5n + 3 is even.

Solution:

Suppose n is odd.	given
Therefore $n = 2k + 1$ for some integer k	definition of odd integer
Therefore $5n + 3 = 5(2k + 1) + 3$	substitution of $2k + 1$ for n
Therefore $5n + 3 = 10k + 8$	algebra
Therefore $5n + 3 = 2(5k + 4)$	algebra
Therefore $5n + 3$ is even	$5n + 3$ is an integer multiple of 2

We give a proof using two columns, one for statements and one for reasons.

Extra exercises

Prove that the square of every even integer ends in 0, 4, or 6.

Solution:
Every even integer n can be written as $n = 10k + r$ where $r = 0, 2, 4, 6, 8$. (For example, $34 = 10 \cdot 3 + 4$ and $6 = 6 \cdot 0 + 6$.)
Examine each of these five cases separately:

$n = 10k + 0$: $n^2 = 100k^2$, which ends in 0 (it is a multiple of 10)
$n = 10k + 2$: $n^2 = 100k^2 + 40k + 4 = 10(10k^2 + 4k) + 4$ and hence ends in 4
$n = 10k + 4$: $n^2 = 100k^2 + 80k + 16 = 10(10k^2 + 8k + 1) + 6$ and hence ends in 6
$n = 10k + 6$: $n^2 = 100k^2 + 120k + 36 = 10(10k^2 + 12k + 3) + 6$ and hence ends in 6
$n = 10k + 8$: $n^2 = 100k^2 + 160k + 64 = 10(10k^2 + 16k + 6) + 4$ and hence ends in 4.

Extra exercises

Suppose a and b are integers such that $2a = b^2 + 3$. Prove that a is the sum of three squares.

Solution:

We first observe that $b^2 + 3$ is even — it is equal to $2a$. Because 3 is odd, b^2 must also be odd. Therefore b must be odd and we can write $b = 2n + 1$. Therefore

$$2a = b^2 + 3 = (2n+1)^2 + 3 = 4n^2 + 4n + 4.$$

Dividing by 2, we have $a = 2n^2 + 2n + 2$. We need to examine $2n^2 + 2n + 2$ and try to write it as the sum of three squares.

If we write $2n^2$ as $n^2 + n^2$, we have

$$a = n^2 + n^2 + 2n + 2.$$

This gives two squares, but the remaining terms, $2n + 2$, does not appear to be a square. But if we write this as $a = n^2 + n^2 + 2n + 1 + 1$ we obtain a as sum of three squares:

$$a = n^2 + (n^2 + 2n + 1) + 1 = n^2 + (n + 1)^2 + 1.$$

Key Terms and Results

TERMS

proposition: a statement that is true or false

propositional variable: a variable that represents a proposition

truth value: true or false

$\neg p$ (negation of p): the proposition with truth value opposite to the truth value of p

logical operators: operators used to combine propositions

compound proposition: a proposition constructed by combining propositions using logical operators

truth table: a table displaying all possible truth values of propositions

$p \vee q$ (disjunction of p and q): the proposition "p or q," which is true if and only if at least one of p and q is true

$p \wedge q$ (conjunction of p and q): the proposition "p and q," which is true if and only if both p and q are true

$p \oplus q$ (exclusive or of p and q): the proposition "p *XOR* q," which is true when exactly one of p and q is true

$p \rightarrow q$ (p implies q): the proposition "if p, then q," which is false if and only if p is true and q is false

converse of $p \rightarrow q$: the conditional statement $q \rightarrow p$

contrapositive of $p \rightarrow q$: the conditional statement $\neg q \rightarrow \neg p$

inverse of $p \rightarrow q$: the conditional statement $\neg p \rightarrow \neg q$

$p \leftrightarrow q$ (biconditional): the proposition "p if and only if q," which is true if and only if p and q have the same truth value

bit: either a 0 or a 1

Boolean variable: a variable that has a value of 0 or 1

bit operation: an operation on a bit or bits

bit string: a list of bits

bitwise operations: operations on bit strings that operate on each bit in one string and the corresponding bit in the other string

logic gate: a logic element that performs a logical operation on one or more bits to produce an output bit

logic circuit: a switching circuit made up of logic gates that produces one or more output bits

tautology: a compound proposition that is always true

contradiction: a compound proposition that is always false

contingency: a compound proposition that is sometimes true and sometimes false

consistent compound propositions: compound propositions for which there is an assignment of truth values to the variables that makes all these propositions true

satisfiable compound proposition: a compound proposition for which there is an assignment of truth values to its variables that makes it true

logically equivalent compound propositions: compound propositions that always have the same truth values

predicate: part of a sentence that attributes a property to the subject

propositional function: a statement containing one or more variables that becomes a proposition when each of its variables is assigned a value or is bound by a quantifier

domain (or universe) of discourse: the values a variable in a propositional function may take

$\exists x\, P(x)$ (existential quantification of $P(x)$): the proposition that is true if and only if there exists an x in the domain such that $P(x)$ is true

$\forall x P(x)$ (universal quantification of $P(x)$): the proposition that is true if and only if $P(x)$ is true for every x in the domain

logically equivalent expressions: expressions that have the same truth value no matter which propositional functions and domains are used

free variable: a variable not bound in a propositional function

bound variable: a variable that is quantified

scope of a quantifier: portion of a statement where the quantifier binds its variable

argument: a sequence of statements

argument form: a sequence of compound propositions involving propositional variables

premise: a statement, in an argument, or argument form, other than the final one

conclusion: the final statement in an argument or argument form

valid argument form: a sequence of compound propositions involving propositional variables where the truth of all the premises implies the truth of the conclusion

valid argument: an argument with a valid argument form

rule of inference: a valid argument form that can be used in the demonstration that arguments are valid

fallacy: an invalid argument form often used incorrectly as a rule of inference (or sometimes, more generally, an incorrect argument)

circular reasoning or begging the question: reasoning where one or more steps are based on the truth of the statement being proved

theorem: a mathematical assertion that can be shown to be true

conjecture: a mathematical assertion proposed to be true, but that has not been proved

proof: a demonstration that a theorem is true

axiom: a statement that is assumed to be true and that can be used as a basis for proving theorems

lemma: a theorem used to prove other theorems

corollary: a proposition that can be proved as a consequence of a theorem that has just been proved

vacuous proof: a proof that $p \to q$ is true based on the fact that p is false

trivial proof: a proof that $p \to q$ is true based on the fact that q is true

direct proof: a proof that $p \to q$ is true that proceeds by showing that q must be true when p is true

proof by contraposition: a proof that $p \to q$ is true that proceeds by showing that p must be false when q is false

proof by contradiction: a proof that p is true based on the truth of the conditional statement $\neg p \to q$, where q is a contradiction

exhaustive proof: a proof that establishes a result by checking a list of all possible cases

proof by cases: a proof broken into separate cases, where these cases cover all possibilities

without loss of generality: an assumption in a proof that makes it possible to prove a theorem by reducing the number of cases to consider in the proof

counterexample: an element x such that $P(x)$ is false

constructive existence proof: a proof that an element with a specified property exists that explicitly finds such an element

nonconstructive existence proof: a proof that an element with a specified property exists that does not explicitly find such an element

rational number: a number that can be expressed as the ratio of two integers p and q such that $q \neq 0$

uniqueness proof: a proof that there is exactly one element satisfying a specified property

Basic Structures: Sets, Functions, Sequences, Sums, and Matrices

Chapter 2

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Chapter Summary

Sets

- The Language of Sets
- Set Operations
- Set Identities

Functions

- Types of Functions
- Operations on Functions
- Computability

Sequences and Summations

- Types of Sequences
- Summation Formulae

Set Cardinality

- Countable Sets

Matrices

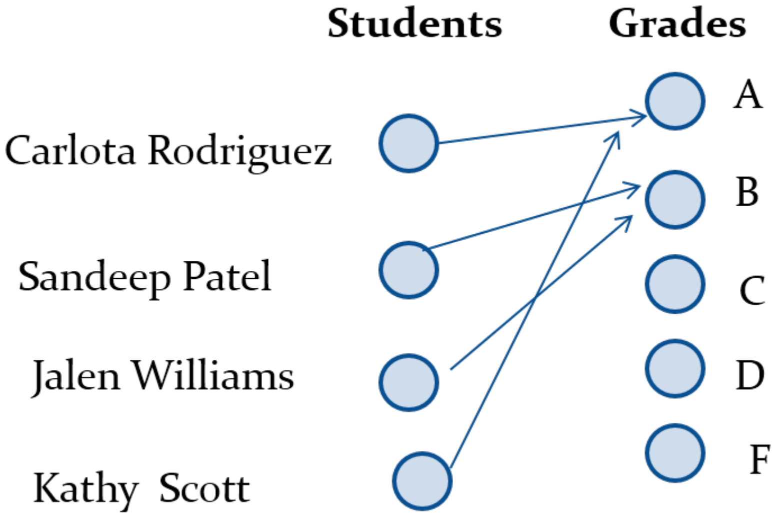
- Matrix Arithmetic

2.3.1 Functions [1]

Definition: Let A and B be nonempty sets. A *function* f from A to B, denoted $f: A \to B$ is an assignment of each element of A to exactly one element of B.

We write $f(a) = b$ if b is the unique element of B assigned by the function f to the element a of A.

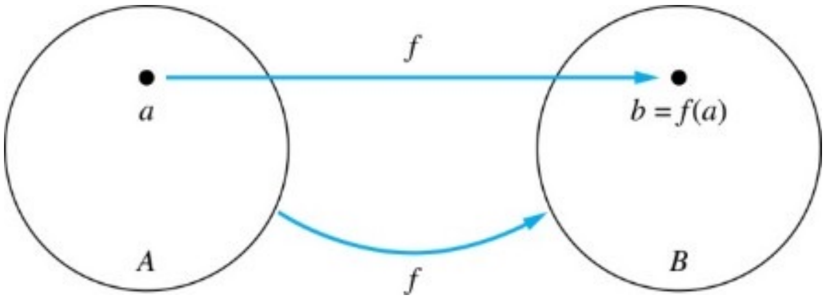
- Functions are sometimes called *mapping* or *transformation*.



Functions $_3$

Given a function f: $A \to B$**:**

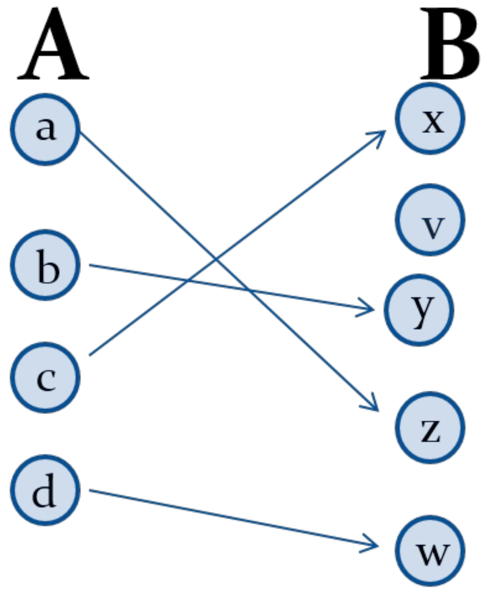
- We say f *maps* A to B *or* f is a *mapping* from A to B.
- A is called the *domain* of f.
- B is called the *codomain* of f.
- If $f(a) = b$,
  - then b is called the *image* of a under f.
  - a is called the *preimage* of b.



- The range of f is the set of all images of points in **A** under f. We denote it by $f(\boldsymbol{A})$.
- Two functions are *equal* when they have the same domain, the same codomain and map each element of the domain to the same element of the codomain.

One-to-One

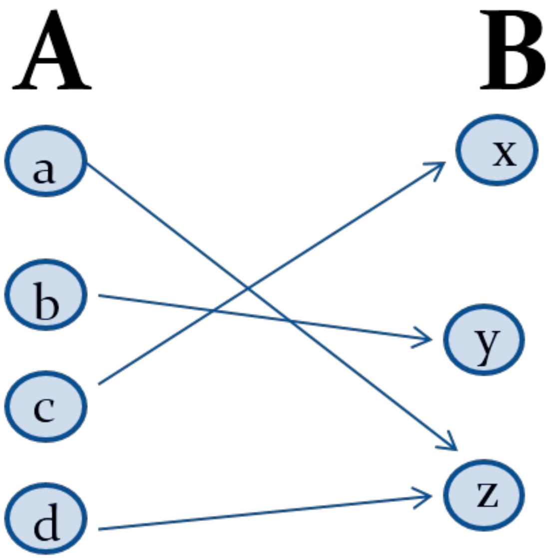
Some functions never assign the same value to two different domain elements. These functions are said to be one-to-one.



Definition: A function f is said to be *one-to-one* , or *an injection*, if and only if $f(a) = f(b)$ implies that $a = b$ for all a and b in the domain of f. A function is said to be injective if it is one-to-one.

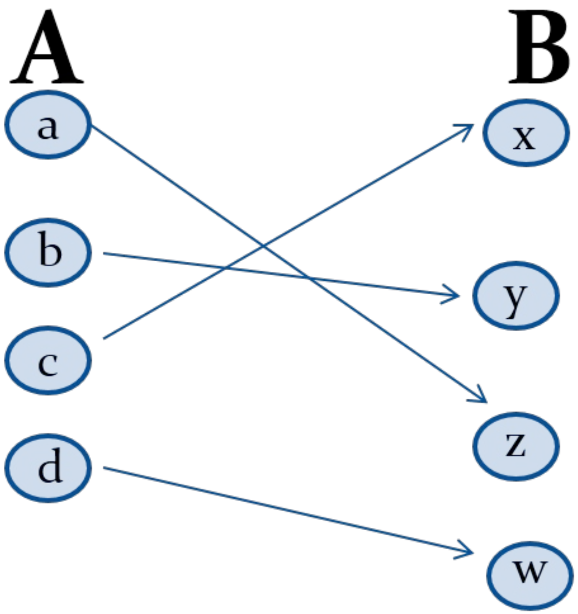
Onto

Definition: A function *f* from *A* to *B* is called *onto* or *surjective*, if and only if for every element $b \in B$ there is an element $a \in A$ with $f(a) = b$.
A function *f* is called a *surjection* if it is *onto*.

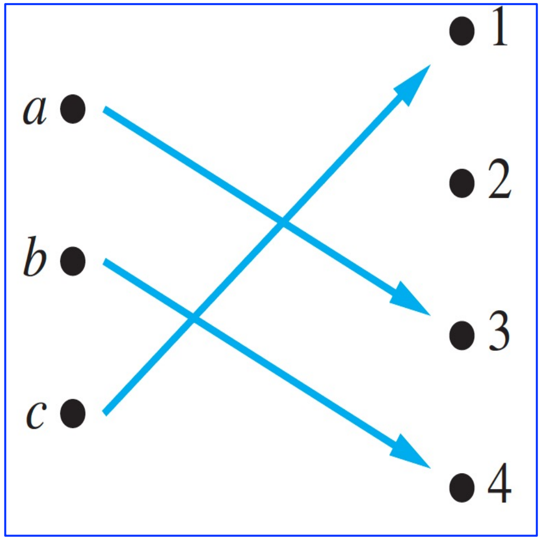


Bijections

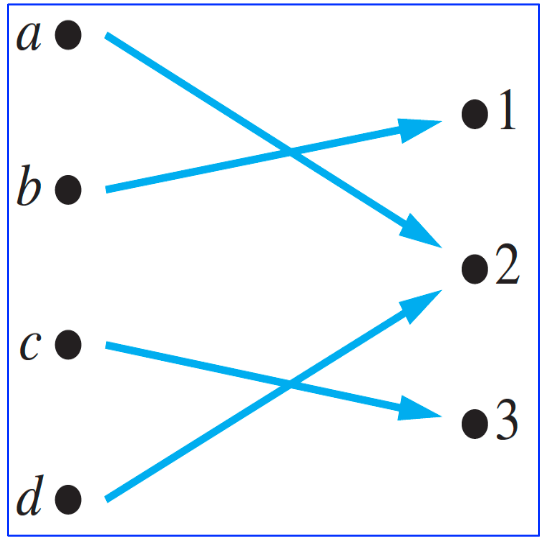
Definition: A function f is a *one-to-one correspondence*, or a *bijection*, if it is both one-to-one and onto (surjective and injective).



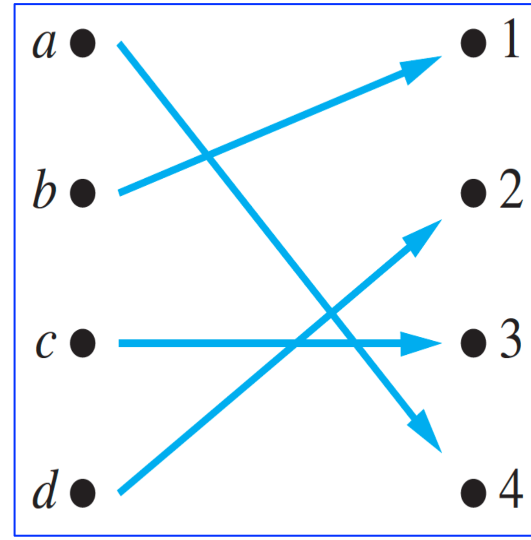
2.3.2 One-to-One and Onto Functions



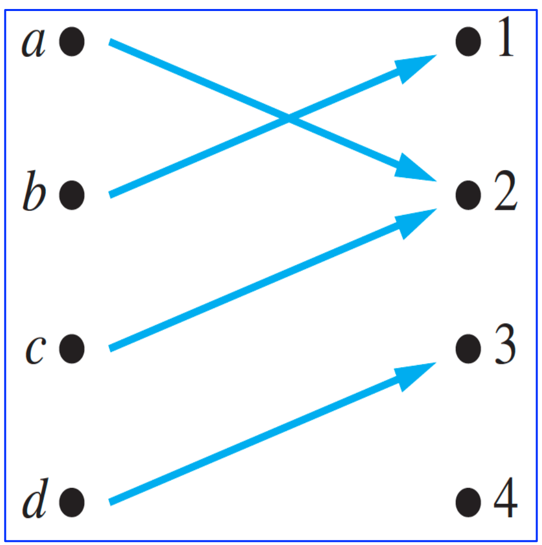
One-to-one, not onto



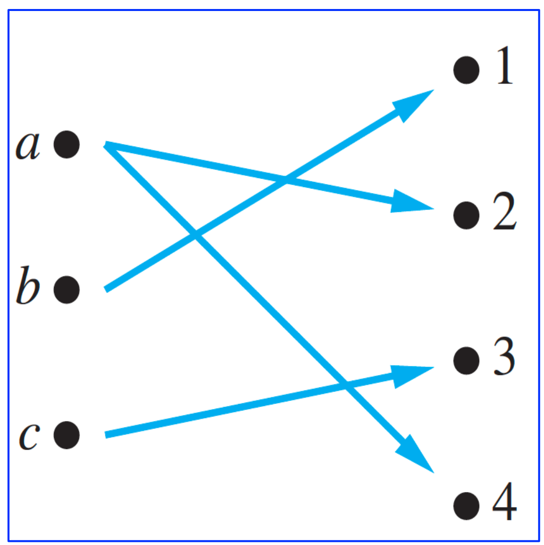
Onto, not one-to-one



One-to-one and onto



Neither one-to-one nor onto



Not a function

Showing that f is one-to-one or onto 1

Suppose that $f : A \to B$.

- *To show that f is injective*

    Show that if $f(x) = f(y)$ for arbitrary $x, y \in A$, then $x = y$.

- *To show that f is not injective*

    Find particular elements $x, y \in A$ such that $x \neq y$ and $f(x) = f(y)$.

- *To show that f is surjective*

    Consider an arbitrary element $y \in B$ and find an element $x \in A$ such that $f(x) = y$.

- *To show that f is not surjective*

    Find a particular $y \in B$ such that $f(x) \neq y$ for all $x \in A$.

Composition Questions 2

Example 2: Let f and g be functions from the set of integers to the set of integers defined by

$$f(x) = 2x + 3 \quad \text{and} \quad g(x) = 3x + 2.$$

What is the composition of *f* and *g*, and also the composition of *g* and *f* ?

Solution:

$$f \circ g\ (x) = f\big(g(x)\big) = f(3x + 2) = 2(3x + 2) + 3 = 6x + 7$$
$$g \circ f\ (x) = g\big(f(x)\big) = g(3x + 2) = 3(3x + 2) + 2 = 6x + 11$$

2.3.5 Some Important Functions

- The *floor* function, denoted $f(x) = \lfloor x \rfloor$ is the largest integer less than or equal to *x*.
- The *ceiling* function, denoted $f(x)\lceil x \rceil$ is the smallest integer greater than or equal to *x.*

Example:

$$\lceil 3.5 \rceil = 4 \qquad \lfloor 3.5 \rfloor = 3$$

$$\lceil -1.5 \rceil = -1 \qquad \lfloor -1.5 \rfloor = -2$$

Arithmetic Progression

Definition: A arithmetic progression is a sequence of the form: $a, a+d, a+2d, \ldots, a+nd, \ldots$

where the *initial term* *a* and the *common difference* *d* are real numbers.

Examples :

1. Let $a = -1$ and $d = 4$:

$$\{s_n\} = \{s_0, s_1, s_2, s_3, s_4, \ldots\} = \{-1, 3, 7, 11\}$$

2. Let $a = 7$ and $d = -3$:

$$\{t_n\} = \{t_0, t_1, t_2, t_3, t_4, \ldots\} = \{7, 4, 1, -2, -5, \ldots\}$$

3. Let $a = 1$ and $d = 2$:

$$\{u_n\} = \{u_0, u_1, u_2, u_3, u_4, \ldots\} = \{1, 3, 5, 7, 9, \ldots\}$$

2.5.1 Cardinality

Definition: The *cardinality* of a set A is equal to the cardinality of a set B, denoted

$$|A| = |B|,$$

if and only if there is a one-to-one correspondence (*i.e.*, a bijection) from A to B.

If there is a one-to-one function (*i.e.*, an injection) from A to B, the cardinality of A is less than or the same as the cardinality of B and we write $|A| \leq |B|$.

When $|A| \leq |B|$ and A and B have different cardinality, we say that the cardinality of A is less than the cardinality of B and write $|A| < |B|$.

2.5.2 Countable Sets

Definition: A set that is either finite or has the same cardinality as the set of positive integers (**Z**$^+$) is called *countable*. A set that is not countable is *uncountable*.

The set of real numbers **R** is an uncountable set.

When an infinite set is countable (*countably infinite*) its cardinality is $\aleph_0$ (where $\aleph$ is aleph, the 1st letter of the Hebrew alphabet). We write $|S| = \aleph_0$ and say that S has cardinality "aleph null."

Showing that a Set is Countable [1]

An infinite set is countable if and only if it is possible to list the elements of the set in a sequence (indexed by the positive integers).

The reason for this is that a one-to-one correspondence f from the set of positive integers to a set S can be expressed in terms of a sequence $a_1, a_2, \ldots, a_n, \ldots$ where $a_1 = f(1)$, $a_2 = f(2), \ldots, a_n = f(n), \ldots$

Showing that a Set is Countable [2]

Show that the set of positive even integers E is countable set.

To show that a set is countable, we will exhibit a one-to-one correspondence between that set and the set of positive integers.

Solution: $f(x) = 2x.$

$$\begin{array}{cccccc} 1 & 2 & 3 & 4 & 5 & 6 \\ \updownarrow & \updownarrow & \updownarrow & \updownarrow & \updownarrow & \updownarrow \\ 2 & 4 & 6 & 8 & 10 & 12 \end{array}$$

Then f is a bijection from **N** to E since f is both one-to-one and onto.

To show that it is one-to-one, suppose that $f(n) = f(m)$. Then $2n = 2m$, and so $n = m$.

To see that it is onto, suppose that t is an even positive integer. Then $t = 2k$ for some positive integer k and $f(k) = t$.

Showing that a Set is Countable 3

Show that the set of integers **Z** is countable.

Solution: Can list in a sequence:

0, 1, − 1, 2, − 2, 3, − 3 ,……….

Or can define a bijection from **N** to **Z**:

- When n is even: $f(n) = n/2$
- When n is odd: $f(\mathrm{n}) = -(n-1)/2$

The Positive Rational Numbers are Countable[1]

Definition: A *rational number* can be expressed as the ratio of two integers p and q such that $q \neq 0$.

- 3/4 is a rational number
- $\sqrt{2}$ is not a rational number.

Show that the positive rational numbers are countable.

Solution: The positive rational numbers are countable since they can be arranged in a sequence:

$$r_1, r_2, r_3, \ldots$$

The next slide shows how this is done.

Results about cardinality

Schroder-Bernstein theorem:

If A and B are sets with $|A| \le |B|$ and $|B| \le |A|$, then $|A| = |B|$. In other words, if there are one-to-one functions f from A to B and g from B to A, then there is a one-to-one correspondence between A and B.

Show that the $|(0, 1)| = |(0, 1]|$.

$f(x) = x$ is a one-to-one function from $(0, 1)$ to $(0, 1]$.

$g(x) = x/2$ is a one-to-one function from $(0, 1]$ to $(0, 1/2] \subset (0, 1)$.

Matrix

Definition: A *matrix* is a rectangular array of numbers. A matrix with m rows and n columns is called an $m \times n$ matrix.

- The plural of matrix is *matrices*.
- A matrix with the same number of rows as columns is called *square*.
- Two matrices are *equal* if they have the same number of rows and the same number of columns and the corresponding entries in every position are equal.

3×2 matrix $\begin{bmatrix} 1 & 1 \\ 0 & 2 \\ 1 & 3 \end{bmatrix}$

Key Terms and Results

TERMS

set: an unordered collection of distinct objects

axiom: a basic assumption of a theory

paradox: a logical inconsistency

element, member of a set: an object in a set

roster method: a method that describes a set by listing its elements

set builder notation: the notation that describes a set by stating a property an element must have to be a member

multiset: an unordered collection of objects where objects can occur multiple times

$\emptyset$ (empty set, null set): the set with no members

universal set: the set containing all objects under consideration

Venn diagram: a graphical representation of a set or sets

$S = T$ (set equality): S and T have the same elements

$S \subseteq T$ (S is a subset of T): every element of S is also an element of T

$S \subset T$ (S is a proper subset of T): S is a subset of T and $S \neq T$

finite set: a set with n elements, where n is a nonnegative integer

infinite set: a set that is not finite

$|S|$ (the cardinality of S): the number of elements in S

$P(S)$ (the power set of S): the set of all subsets of S

$A \cup B$ (the union of A and B): the set containing those elements that are in at least one of A and B

$A \cap B$ (the intersection of A and B): the set containing those elements that are in both A and B.

$A - B$ (the difference of A and B): the set containing those elements that are in A but not in B

$\overline{A}$ (the complement of A): the set of elements in the universal set that are not in A

$A \oplus B$ (the symmetric difference of A and B): the set containing those elements in exactly one of A and B

membership table: a table displaying the membership of elements in sets

function from A to B: an assignment of exactly one element of B to each element of A

domain of f: the set A, where f is a function from A to B

codomain of f: the set B, where f is a function from A to B

b is the image of a under f: $b = f(a)$

a is a preimage of b under f: $f(a) = b$

range of f: the set of images of f

onto function, surjection: a function from A to B such that every element of B is the image of some element in A

one-to-one function, injection: a function such that the images of elements in its domain are distinct

one-to-one correspondence, bijection: a function that is both one-to-one and onto

inverse of f: the function that reverses the correspondence given by f (when f is a bijection)

$f \circ g$ (composition of f and g): the function that assigns $f(g(x))$ to x

$\lfloor x \rfloor$ (floor function): the largest integer not exceeding x

$\lceil x \rceil$ (ceiling function): the smallest integer greater than or equal to x

partial function: an assignment to each element in a subset of the domain a unique element in the codomain

sequence: a function with domain that is a subset of the set of integers

geometric progression: a sequence of the form $a, ar, ar^2, \ldots,$ where a and r are real numbers

arithmetic progression: a sequence of the form $a, a+d, a+2d, \ldots,$ where a and d are real numbers

string: a finite sequence

empty string: a string of length zero

recurrence relation: a equation that expresses the nth term a_n of a sequence in terms of one or more of the previous terms of the sequence for all integers n greater than a particular integer

$\sum_{i=1}^{n} a_i$: the sum $a_1 + a_2 + \cdots + a_n$

$\prod_{i=1}^{n} a_i$: the product $a_1 a_2 \cdots a_n$

cardinality: two sets A and B have the same cardinality if there is a one-to-one correspondence from A to B

countable set: a set that either is finite or can be placed in one-to-one correspondence with the set of positive integers

uncountable set: a set that is not countable

$\aleph_0$ (aleph null): the cardinality of a countable set

$\mathfrak{c}$: the cardinality of the set of real numbers

Cantor diagonalization argument: a proof technique used to show that the set of real numbers is uncountable

computable function: a function for which there is a computer program in some programming language that finds its values

uncomputable function: a function for which no computer program in a programming language exists that finds its values

continuum hypothesis: the statement that no set A exists such that $\aleph_0 < |A| < \mathfrak{c}$

matrix: a rectangular array of numbers

matrix addition: see page 188

matrix multiplication: see page 189

$\mathbf{I}_n$ (identity matrix of order n): the $n \times n$ matrix that has entries equal to 1 on its diagonal and 0s elsewhere

$\mathbf{A}^t$ (transpose of $\mathbf{A}$): the matrix obtained from $\mathbf{A}$ by interchanging the rows and columns

symmetric matrix: a matrix is symmetric if it equals its transpose

zero–one matrix: a matrix with each entry equal to either 0 or 1

$\mathbf{A} \vee \mathbf{B}$ (the join of $\mathbf{A}$ and $\mathbf{B}$): see page 191

$\mathbf{A} \wedge \mathbf{B}$ (the meet of $\mathbf{A}$ and $\mathbf{B}$): see page 191

$\mathbf{A} \odot \mathbf{B}$ (the Boolean product of $\mathbf{A}$ and $\mathbf{B}$): see page 192

RESULTS

The set identities given in Table 1 in Section 2.2

The summation formulae in Table 2 in Section 2.4

The set of rational numbers is countable.

The set of real numbers is uncountable.

Results

TABLE 1 Set Identities.

Identity	*Name*
$A \cap U = A$ $A \cup \emptyset = A$	Identity laws
$A \cup U = U$ $A \cap \emptyset = \emptyset$	Domination laws
$A \cup A = A$ $A \cap A = A$	Idempotent laws
$\overline{(\overline{A})} = A$	Complementation law
$A \cup B = B \cup A$ $A \cap B = B \cap A$	Commutative laws
$A \cup (B \cup C) = (A \cup B) \cup C$ $A \cap (B \cap C) = (A \cap B) \cap C$	Associative laws
$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$	Distributive laws
$\overline{A \cap B} = \overline{A} \cup \overline{B}$ $\overline{A \cup B} = \overline{A} \cap \overline{B}$	De Morgan's laws
$A \cup (A \cap B) = A$ $A \cap (A \cup B) = A$	Absorption laws
$A \cup \overline{A} = U$ $A \cap \overline{A} = \emptyset$	Complement laws

TABLE 2 Some Useful Summation Formulae.

Sum	*Closed Form*
$\sum_{k=0}^{n} ar^k \ (r \neq 0)$	$\frac{ar^{n+1} - a}{r - 1}, r \neq 1$
$\sum_{k=1}^{n} k$	$\frac{n(n+1)}{2}$
$\sum_{k=1}^{n} k^2$	$\frac{n(n+1)(2n+1)}{6}$
$\sum_{k=1}^{n} k^3$	$\frac{n^2(n+1)^2}{4}$
$\sum_{k=0}^{\infty} x^k, \lvert x \rvert < 1$	$\frac{1}{1-x}$
$\sum_{k=1}^{\infty} kx^{k-1}, \lvert x \rvert < 1$	$\frac{1}{(1-x)^2}$

- The set of rational numbers is countable.
- The set of real numbers is uncountable.

Counting

Chapter 6

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Chapter Summary

6.1 The Basics of Counting

6.2 The Pigeonhole Principle

6.3 Permutations and Combinations

6.4 Binomial Coefficients and Identities

6.5 Generalized Permutations and Combinations

6.6 Generating Permutations and Combinations

Section Summary

The Product Rule

The Sum Rule

The Subtraction Rule

The Division Rule

Tree Diagrams

6.2.1 The Pigeonhole Principle

If a flock of 20 pigeons roosts in a set of 19 pigeonholes, one of the pigeonholes must have more than 1 pigeon.

THEOREM 1 If k is a positive integer and $k + 1$ objects are placed into k boxes, then at least one box contains two or more objects.

Proof: We use a proof by contraposition. Suppose <u>none of the k boxes</u> has <u>more than one object</u>. Then the total number of objects would be at <u>most k</u>. This <u>contradicts</u> the statement that we have <u>$k + 1$ objects</u>.

Poker

Example:(a) How many cards must be selected from a standard deck of 52 cards to guarantee that at least three cards of the same suit are chosen?

The smallest integer N such that $\lceil N/4 \rceil \geq 3$ (Four suit)is $N = 2 \cdot 4 + 1 = 9$.

(b) How many must be selected to guarantee that at least three hearts are selected?

A deck contains 13 hearts and 39 cards which are not hearts.

So, when we select 42 cards, we must have at least three hearts.

Four suit

Arrange

Permutations and Combinations

$$A_n^m = n(n-1)(n-2)\cdots(n-m+1)$$
$$A_n^m = \frac{n!}{(n-m)!}$$
$$C_n^m = \frac{A_n^m}{m!} = \frac{n(n-1)\cdots(n-m+1)}{m!}$$
$$C_n^m = \frac{n!}{m!(n-m)!}$$
$$C_{n+1}^m = C_n^m + C_n^{m-1}$$
$$C_n^m = C_n^{n-m}$$

Summarizing the Formulas for Counting Permutations and Combinations with and without Repetition

Type	***Repetition Allowed?***	***Formula***
r-permutations	No	$\frac{n!}{(n-r)!}$
r-combinations	No	$\frac{n!}{r!(n-r)!}$
r-permutations	Yes	n^r
r-combinations	Yes	$\frac{(n+r-1)!}{r!(n-1)!}$

6.5.4 Permutations with Indistinguishable Objects

Example: How many different strings can be made by reordering the letters of the word *SUCCESS*.

Solution: There are seven possible positions for the three Ss, two Cs, one U, and one E.

- The three Ss can be placed in $C(7,3)$ different ways, leaving four positions free.
- The two Cs can be placed in $C(4,2)$ different ways, leaving two positions free.
- The U can be placed in $C(2,1)$ different ways, leaving one position free.
- The E can be placed in $C(1,1)$ way.

By the product rule, the number of different strings is:

$$C(7,3)C(4,2)C(2,1)C(1,1) = \frac{7!}{3!4!} \cdot \frac{4!}{2!2!} \cdot \frac{2!}{1!1!} \cdot \frac{1!}{3!2!1!1!} = 420.$$

Summary: Objects into Boxes

distinguishable boxes

- distinguishable objects: The number of ways to distribute n distinguishable objects into k distinguishable boxes so that n_i objects are placed into box i, i = 1, 2,..., k, equals $\frac{n!}{n_1!n_2!\cdots n_k!}$.
- indistinguishable objects: This means that there are C(n + r − 1, n − 1) ways to place r indistinguishable objects into n distinguishable boxes.

indistinguishable boxes

- No simple closed formula exists for this number.

6.6.2 Generating Permutations

Any set with n elements can be placed in one-to-one correspondence with the set $\{1, 2, 3, \ldots, n\}$.

We can list the permutations of any set of n elements by generating the permutations of the n smallest positive integers ($\{1, 2, 3, \ldots, n\}$),

and then replacing these integers with the corresponding elements.

We will describe one of these that is based on the lexicographic (or dictionary) ordering of the set of permutations of $\{1, 2, 3, \ldots, n\}$.

Permutation in Lexicographic Order

In this ordering, the permutation $a_1a_2\cdots a_n$ precedes the permutation of $b_1b_2\cdots b_n$, if for some k, with $1 \leq k \leq n$, $a_1 = b_1$, $a_2 = b_2$,..., $a_{k-1} = b_{k-1}$, and $a_k < b_k$.

EXAMPLE: The permutation 23415 of the set {1, 2, 3, 4, 5} precedes the permutation 23514.

Similarly, the permutation 41532 precedes 52143.

Permutation in Lexicographic Order

EXAMPLE: What is the next permutation in lexicographic order after 362541?

362541

Solution:

364125

The last pair of integers a_j and a_{j+1} where $a_j < a_{j+1}$ is $a_3 = 2$ and $a_4 = 5$.

The least integer to the right of 2 that is greater than 2 in the permutation is $a_5 = 4$.

Hence, 4 is placed in the third position.

Then the integers 2, 5, and 1 are placed in order in the last three positions, giving 125 as the last three positions of the permutation.

Hence, the next permutation is 364125.

Key Terms and Results

TERMS

combinatorics: the study of arrangements of objects

enumeration: the counting of arrangements of objects

tree diagram: a diagram made up of a root, branches leaving the root, and other branches leaving some of the endpoints of branches

permutation: an ordered arrangement of the elements of a set

***r*-permutation:** an ordered arrangement of r elements of a set

$P(n,r)$: the number of r-permutations of a set with n elements

***r*-combination:** an unordered selection of r elements of a set

$C(n,r)$: the number of r-combinations of a set with n elements

binomial coefficient $\binom{n}{r}$: also the number of r-combinations of a set with n elements

combinatorial proof: a proof that uses counting arguments rather than algebraic manipulation to prove a result

Pascal's triangle: a representation of the binomial coefficients where the ith row of the triangle contains $\binom{i}{j}$ for $j = 0, 1, 2, \ldots, i$

$S(n,j)$: the Stirling number of the second kind denoting the number of ways to distribute n distinguishable objects into j indistinguishable boxes so that no box is empty

Key Terms and Results

RESULTS

product rule for counting: The number of ways to do a procedure that consists of two tasks is the product of the number of ways to do the first task and the number of ways to do the second task after the first task has been done.

product rule for sets: The number of elements in the Cartesian product of finite sets is the product of the number of elements in each set.

sum rule for counting: The number of ways to do a task in one of two ways is the sum of the number of ways to do these tasks if they cannot be done simultaneously.

sum rule for sets: The number of elements in the union of pairwise disjoint finite sets is the sum of the numbers of elements in these sets.

subtraction rule for counting or **inclusion–exclusion for sets:** If a task can be done in either n_1 ways or n_2 ways, then the number of ways to do the task is $n_1 + n_2$ minus the number of ways to do the task that are common to the two different ways.

subtraction rule or **inclusion–exclusion for sets:** The number of elements in the union of two sets is the sum of the number of elements in these sets minus the number of elements in their intersection.

division rule for counting: There are n/d ways to do a task if it can be done using a procedure that can be carried out in n ways, and for every way w, exactly d of the n ways correspond to way w.

division rule for sets: Suppose that a finite set A is the union of n disjoint subsets each with d elements. Then $n = |A|/d$.

the pigeonhole principle: When more than k objects are placed in k boxes, there must be a box containing more than one object.

the generalized pigeonhole principle: When N objects are placed in k boxes, there must be a box containing at least $\lceil N/k \rceil$ objects.

$$P(n, r) = \frac{n!}{(n-r)!}$$

$$C(n, r) = \binom{n}{r} = \frac{n!}{r!(n-r)!}$$

Pascal's identity: $\binom{n+1}{k} = \binom{n}{k-1} + \binom{n}{k}$

the binomial theorem: $(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^{n-k} y^k$

There are n^r r-permutations of a set with n elements when repetition is allowed.

There are $C(n + r - 1, r)$ r-combinations of a set with n elements when repetition is allowed.

There are $n!/(n_1!\, n_2! \cdots n_k!)$ permutations of n objects of k types where there are n_i indistinguishable objects of type i for $i = 1, 2, 3, \ldots, k$.

the algorithm for generating the permutations of the set $\{1, 2, \ldots, n\}$

Relations

Chapter 9

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Chapter Summary

9.1 Relations and Their Properties

- Reflexive, Symmetric, Antisymmetric, and Transitive

9.2 n-ary Relations and Their Applications

- domains, degree
- Databases and Relations： Records, Tables, Primary key, Composite key
- projection,Join operation
- Association Rules from Data Mining

9.3 Representing Relations

- Representing Relations using Matrices or Digraphs

9.4 Closures of Relations

- Reflexive Closure, Symmetric Closure, Transitive Closure
- Connectivity Relation $R*$, Warshall’s Algorithm

9.5 Equivalence Relations

- Equivalence Classes and Partitions

9.6 Partial Orderings

- Partial Orderings and Partially-ordered Sets
- Lexicographic Orderings
- Hasse Diagrams

Reflexive Relations

DEFINITION 3: A relation R on a set A is called reflexive if (a, a) ∈ R for every element a ∈ A.

R is *reflexive* iff (*a*,*a*) ∈ *R* for every element *a* ∈ A.

Written symbolically, R is reflexive if and only if

$$\forall x\left[\mathrm{x} \in U \rightarrow (x, x) \in R\right]$$

$$\forall a((a, a) \in R)$$

If *A* = ∅ then the empty relation is reflexive vacuously.
That is the empty relation on an empty set is reflexive!

Symmetric and Antisymmetric Relations

DEFINITION 4

A relation R on a set A is called symmetric if (b, a) ∈ R whenever (a, b) ∈ R, for all a, b ∈ A.

$$\forall x \forall y \left[\left(x, y \right) \in R \rightarrow \left(y, x \right) \in R \right]$$

A relation R on a set A such that for all a, b ∈ A, if (a, b) ∈ R and (b, a) ∈ R, then a = b is called antisymmetric.

$$\forall x \forall y \left[\left(x, y \right) \in R \wedge \left(y, x \right) \in R \rightarrow x = y \right]$$

Transitive Relations

Let R be the relation consisting of all pairs (x, y) of students at your school, where x has taken more credits than y.

If (x, y) and (y, z), then (x, z).

What we have shown is that R has the transitive property, which is defined as follows.

DEFINITION 5: A relation R on a set A is called transitive if whenever (a, b) ∈ R and (b, c) ∈ R, then (a, c) ∈ R, for all a, b, c ∈ A.

$$\forall x \forall y \forall z \left[(x, y) \in R \wedge (y, z) \in R \rightarrow (x, z) \in R \right]$$

9.4.2 Different Types of Closures

If R is a relation on a set A, it may or may not have some property P, such as reflexivity, symmetry, or transitivity.

When R does not enjoy property P, we would like to find the smallest relation S on A with property P that contains R.

DEFINITION 1: If R is a relation on a set A, then the closure of R with respect to P, if it exists, is the relation S on A with property P that contains R and is a subset of every subset of A X A containing R with property P.

Reflexive Closure of a Relation

EXAMPLE: The relation R = {(1, 1), (1, 2), (2, 1), (3, 2)} on the set A = {1, 2, 3} is not reflexive. How can we produce a reflexive relation containing R that is as small as possible?

{(1, 1), (1, 2), (2, 1), (3, 2),(2, 2),(3,3)}

Because this relation contains R, is reflexive, and is contained within every reflexive relation that contains R, it is called the reflexive closure of R.

Reflexive Closure of a Relation

Given a relation R on a set A, the reflexive closure of R can be formed by adding to R all pairs of the form (a, a) with $a \in A$, not already in R.

The addition of these pairs produces a new relation that is reflexive, contains R, and is contained within any reflexive relation containing R.

We see that the reflexive closure of R equals $R \cup \Delta$, where $\Delta = \{(a, a) \mid a \in A\}$ is the diagonal relation on A.

Reflexive Closure of a Relation

EXAMPLE: What is the reflexive closure of the relation $R = \{(a, b) \mid a < b\}$ on the set of integers?

Solutions: The reflexive closure of R is

$$R \cup \Delta = \{(a, b) \mid a < b\} \cup \{(a, a) \mid a \in Z\}$$

$$= \{(a, b) \mid a \leq b\}.$$

Symmetric Closure of a Relation

EXAMPLE: The relation {(1, 1), (1, 2), (2, 2), (2, 3), (3, 1), (3, 2)} on {1, 2, 3} is not symmetric. How can we produce a symmetric relation that is as small as possible and contains R?

{(1, 1), (1, 2), (2, 2), (2, 3), (3, 1), (3, 2), (2,1), (1,3)}

This new relation is symmetric and contains R, it is called the symmetric closure of R.

Symmetric Closure of a Relation

The symmetric closure of a relation R can be constructed by adding all ordered pairs of the form (b, a), where (a, b) is in the relation, that are not already present in R.

The symmetric closure of a relation can be constructed by taking the union of a relation with its inverse; that is, $R \cup R^{-1}$ is the symmetric closure of R, where $R^{-1} = \{(b, a) \mid (a,b) \in R\}$.

Symmetric Closure of a Relation

EXAMPLE: What is the symmetric closure of the relation $R = \{(a, b) \mid a > b\}$ on the set of positive integers?

Solution: The symmetric closure of R is the relation

$$R \cup R^{-1} = \{(a, b) \mid a > b\} \cup \{(b, a) \mid a > b\}$$

$$= \{(a, b) \mid a \neq b\}.$$

Transitive Closure of a Relation ?

Suppose that a relation R is not transitive.

How can we produce a transitive relation that contains R such that this new relation is contained within any transitive relation that contains R?

Can the transitive closure of a relation R be produced by adding all the pairs of the form (a, c), where (a, b) and (b, c) are already in the relation?

Transitive Closure of a Relation ?

Consider the relation R = {(1, 3), (1, 4), (2, 1), (3, 2)} on the set {1, 2, 3, 4}.

This relation is not transitive.

{(1, 3), (1, 4), (2, 1), (3, 2), (1, 2), (2, 3), (2, 4), and (3, 1)}.

{(1, 3), (1, 4), (2, 1),(3, 2), (1, 2), (2, 3),(2, 4), and (3, 1)}.

Adding these pairs does not produce a transitive relation, because here is not contain (3, 4).

This shows that constructing the transitive closure of a relation is more complicated than constructing either the reflexive or symmetric closure.

Transitive Closure of a Relation ?

The rest of this section develops algorithms for constructing transitive closures.

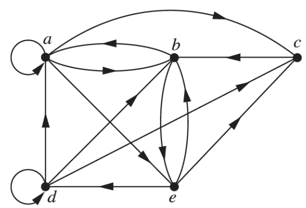
The transitive closure of a relation can be found by adding new ordered pairs that must be present and then repeating this process until no new ordered pairs are needed.

Multi-Closure of a Relation

Symmetric Closure $\rightarrow$ Transitive Closure

9.4.3 Paths in Directed Graphs

Relations by directed graphs helps in the construction of transitive closures.



DEFINITION 2:

A **path** from a to b in the directed graph G is a sequence of edges $(x_0, x_1), (x_1, x_2), (x_2, x_3), \ldots, (x_{n-1}, x_n)$ in G, where n is a nonnegative integer, and $x_0 = a$ and $x_n = b$, that is, a sequence of edges where the terminal vertex of an edge is the same as the initial vertex in the next edge in the path.

This path is denoted by $x_0, x_1, x_2, \ldots, x_{n-1}, x_n$ and has **length** n.

We view the empty set of edges as a path of length zero from a to a.

A path of length $n \geq 1$ that begins and ends at the same vertex is called a **circuit** or **cycle**.

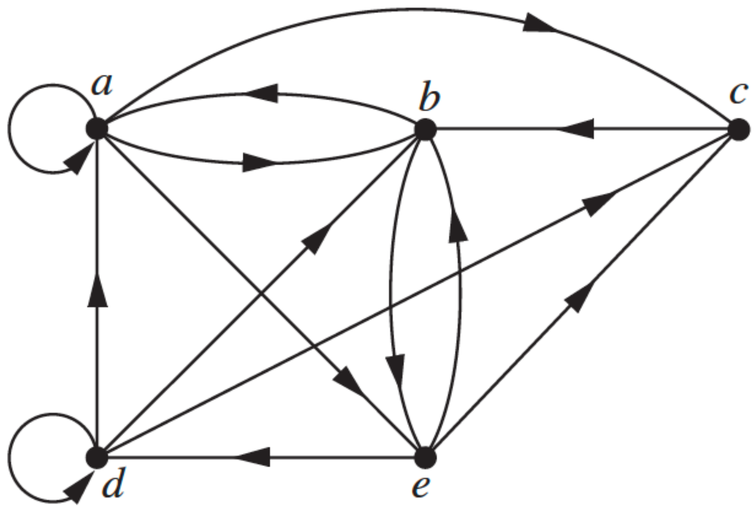
9.4.3 Paths in Directed Graphs

Which of the following are paths in the directed graph ? What are the lengths of those that are paths?

begins and ends at the same vertex

Which of the paths in this list are circuits?

- ✓ ❑ a, b, e, d — 3
- ✗ ❑ a, e, c, d, b
- ✓ ❑ b, a, c, b, a, a, b — 6
- ✓ ❑ d, c — 1
- ✓ ❑ c, b, a — 2
- ✓ ❑ e, b, a, b, a, b, e — 6



Partial Ordering

DEFINITION 1:

A relation R on a set S is called a *partial ordering*, or *partial order,* if it is reflexive, antisymmetric, and transitive.

A set together with a partial ordering R is called a *partially ordered set*, or *poset*, and is denoted by (S, R).

Members of S are called *elements* of the poset.

A represents a 1-ary, but sometimes it can also be n-ary.
For example, an element in A is (a, b)

Partial Ordering

EXAMPLE: Show that the "greater than or equal" relation ($\geq$) is a partial ordering on the set of integers.

- *Reflexivity*: $a \geq a$ for every integer a.
- *Antisymmetry*: If $a \geq b$ and $b \geq a$, then $a = b$.
- *Transitivity*: If $a \geq b$ and $b \geq c$, then $a \geq c$.

It follows that $\geq$ is a partial ordering on the set of integers and $(Z, \geq)$ is a poset.

Recall: a *partially ordered set*, is denoted by (S, R).

Not a Partial Ordering

EXAMPLE: Let R be the relation on the set of people such that xRy if x and y are people and x is older than y. Show that R is not a partial ordering.

- *Reflexivity*: $x \not R x$, no person is older than himself or herself.
- *Antisymmetry*: xRy, then $y \not R x$
- *Transitivity*: if xRy and yRz, then xRz

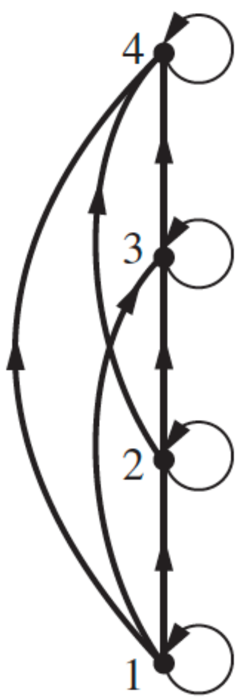
It follows that R is not a partial ordering.

9.6.3 Hasse Diagrams

Many edges in the directed graph for a finite poset do not have to be shown because they must be present.

partial ordering {(a, b) | a ≤ b} on the set {1, 2, 3, 4},

Because this relation is a partial ordering, it is reflexive, and its directed graph has loops at all vertices.



Because a partial ordering is transitive

If we assume that all edges are pointed “upward” , we do not have to show the directions of the edges;

Constructing the Hasse diagram for ({1, 2, 3, 4},≤).

9.6.4 Maximal and Minimal Elements

Elements of posets that have certain extremal properties are important for many applications.

An element of a poset is called maximal if it is not less than any element of the poset.

- a is maximal in the poset $(S, \preccurlyeq)$ if there is no $b \in S$ such that $a \prec b$.

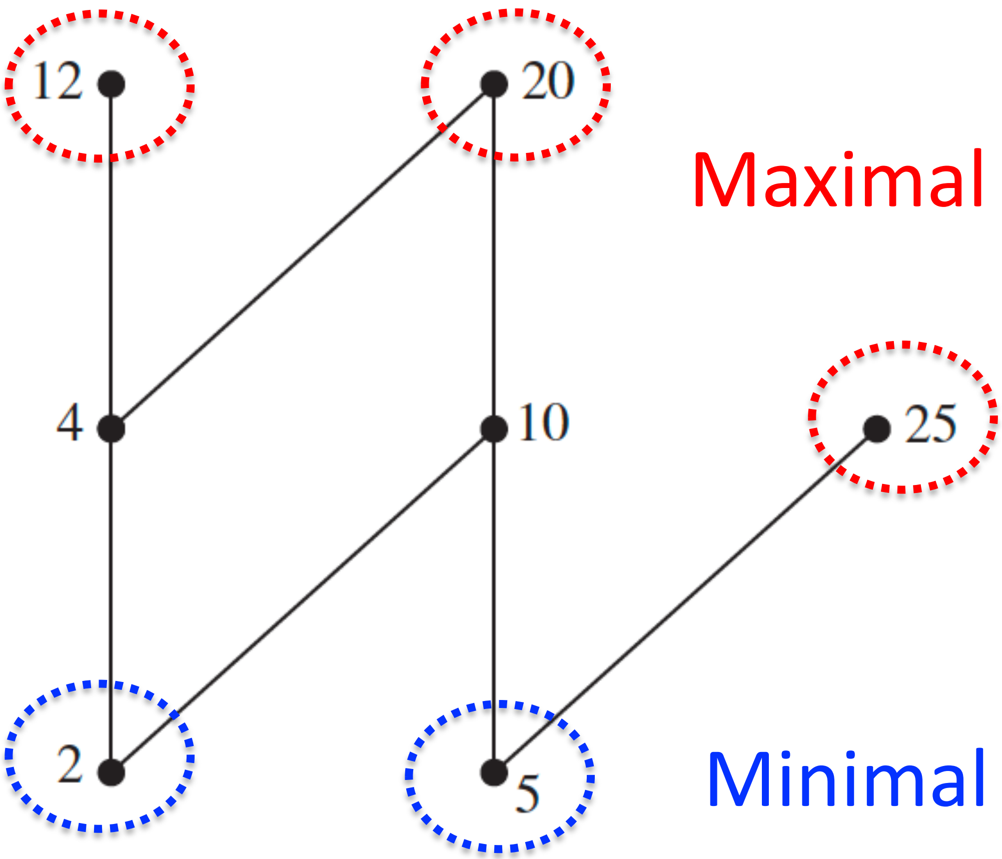
An element of a poset is called minimal if it is not greater than any element of the poset.

- a is minimal if there is no $b \in S$ such that $b \prec a$.

Maximal and minimal elements are easy to spot using a Hasse diagram. They are the "top" and "bottom" elements in the diagram.

Maximal and Minimal Elements

EXAMPLE: Which elements of the poset ({2, 4, 5, 10, 12, 20, 25}, |) are maximal, and which are minimal?



Maximal elements

A poset can have more than one maximal element and more than one minimal element.

Minimal elements

The Hasse diagram of a poset

Greatest and Least Elements

Sometimes there is an element in a poset that is greater than every other element.

Such an element is called the greatest element.

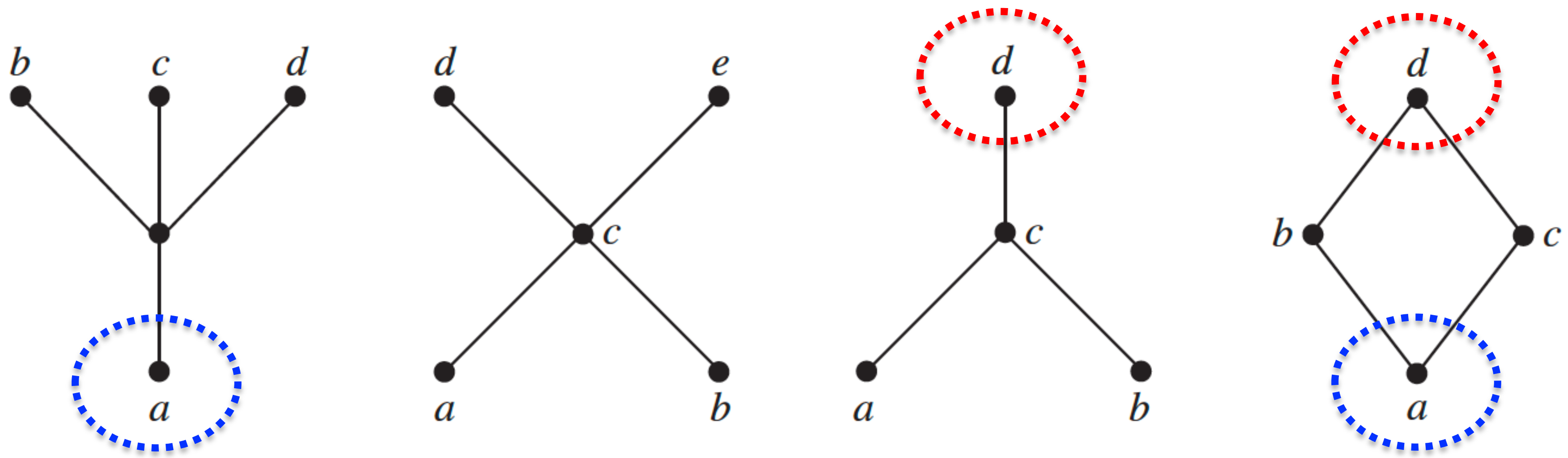
- ❑ a is the greatest element of the poset $(S, \preccurlyeq)$ if $b \preccurlyeq a$ for all $b \in S$.
- ❑ The greatest element is unique when it exists.

Likewise, an element is called the least element if it is less than all the other elements in the poset.

- ❑ a is the least element of $(S, \preccurlyeq)$ if $a \preccurlyeq b$ for all $b \in S$.
- ❑ The least element is unique when it exists.

Greatest and Least Elements

EXAMPLE: Determine whether the posets represented by each of the Hasse diagrams have a greatest element and a least element.



Greatest and Least Elements

EXAMPLE: Let S be a set. Determine whether there is a greatest element and a least element in the poset $(\mathcal{P}(S),\subseteq)$.

- Least element: Empty set, because $\emptyset \subseteq T$ for any subset T of S.
- Greatest element: S, because $T \subseteq S$ whenever T is a subset of S.

EXAMPLE: Is there a greatest element and a least element in the poset (Z+, |)?

- Least element: 1, because $1|n$ whenever n is a positive integer
- Greatest element: No, there is no integer that is divisible by all positive integers,

Upper Bound and Lower Bound

Sometimes it is possible to find an element that is greater than or equal to all the elements in a subset A of a poset $(S, \preccurlyeq)$.

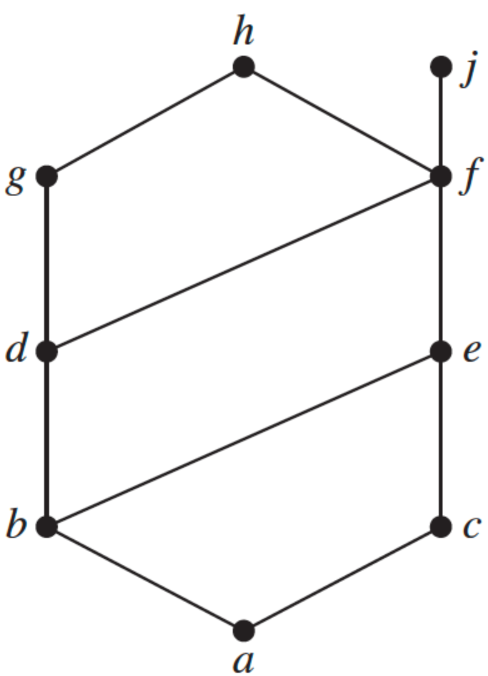
If u is an element of S such that $a \preccurlyeq u$ for all elements $a \in A$, then u is called an upper bound of A.

Likewise, there may be an element less than or equal to all the elements in A.

If l is an element of S such that $l \preccurlyeq a$ for all elements $a \in A$, then l is called a lower bound of A.

Upper Bound and Lower Bound

EXAMPLE: Find the lower and upper bounds of the subsets {a, b, c}, {j, h}, and {a, c, d, f } in the poset with the Hasse diagram.

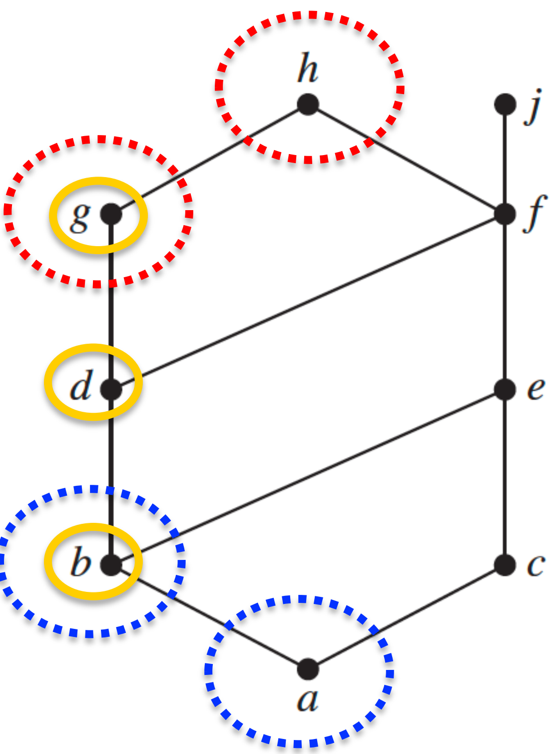


	upper	lower
❑{a, b, c}	e, f, j, and h	a
❑{j, h}	no	a, b, c, d, e, and f
❑{a, c, d, f }	f , h, and j,	a

Least Upper and Greatest Lower Bound

The element x is called the least upper bound of the subset A if x is an upper bound that is less than every other upper bound of A.

Similarly, the element y is called the greatest lower bound of A if y is a lower bound of A and z y whenever z is a lower bound of A.



The upper bounds of {b, d, g} are g and h.

Because g $\prec$ h, g is the least upper bound.

The lower bounds of {b, d, g} are a and b.

Because a $\prec$ b, b is the greatest lower bound.

Key Terms and Results

TERMS

binary relation from A to B: a subset of $A \times B$

relation on A: a binary relation from A to itself (that is, a subset of $A \times A$)

$S \circ R$: composite of R and S

R^{-1}: inverse relation of R

R^n: nth power of R

reflexive: a relation R on A is reflexive if $(a, a) \in R$ for all $a \in A$

symmetric: a relation R on A is symmetric if $(b, a) \in R$ whenever $(a, b) \in R$

antisymmetric: a relation R on A is antisymmetric if $a = b$ whenever $(a, b) \in R$ and $(b, a) \in R$

transitive: a relation R on A is transitive if $(a, b) \in R$ and $(b, c) \in R$ implies that $(a, c) \in R$

n-ary relation on $A_1, A_2, \ldots, A_n$: a subset of $A_1 \times A_2 \times \cdots \times A_n$

relational data model: a model for representing databases using n-ary relations

primary key: a domain of an n-ary relation such that an n-tuple is uniquely determined by its value for this domain

composite key: the Cartesian product of domains of an n-ary relation such that an n-tuple is uniquely determined by its values in these domains

selection operator: a function that selects the n-tuples in an n-ary relation that satisfy a specified condition

projection: a function that produces relations of smaller degree from an n-ary relation by deleting fields

join: a function that combines n-ary relations that agree on certain fields

itemset: a collection of items

count of an itemset: the number of transactions that are supersets of the itemset

frequent itemset: an itemset with frequency greater than or equal to the support threshold

support of an itemset: the frequency of transactions that contain the itemset

association rule: an implication of the form $I \to J$, where I and J are itemsets

support of the association rule $I \to J$: the fraction of transactions that contain both the itemsets I and J

confidence of an association rule: the conditional probability that J is a subset of a transaction given that I is

directed graph or digraph: a set of elements called vertices and ordered pairs of these elements, called edges

loop: an edge of the form (a, a)

closure of a relation R with respect to a property P: the relation S (if it exists) that contains R, has property **P**, and is contained within any relation that contains R and has property **P**

path in a digraph: a sequence of edges $(a, x_1), (x_1, x_2), \ldots, (x_{n-2}, x_{n-1}), (x_{n-1}, b)$ such that the terminal vertex of each edge is the initial vertex of the succeeding edge in the sequence

circuit (or cycle) in a digraph: a path that begins and ends at the same vertex

R^* (connectivity relation): the relation consisting of those ordered pairs (a, b) such that there is a path from a to b

equivalence relation: a reflexive, symmetric, and transitive relation

equivalent: if R is an equivalence relation, a is equivalent to b if aRb

Key Terms and Results

$[a]_R$ (equivalence class of a with respect to R): the set of all elements of A that are equivalent to a

$[a]_m$ (congruence class modulo m): the set of integers congruent to a modulo m

partition of a set S: a collection of pairwise disjoint nonempty subsets that have S as their union

partial ordering: a relation that is reflexive, antisymmetric, and transitive

poset (S, R): a set S and a partial ordering R on this set

comparable: the elements a and b in the poset $(A, \preccurlyeq)$ are comparable if $a \preccurlyeq b$ or $b \preccurlyeq a$

incomparable: elements in a poset that are not comparable

total (or linear) ordering: a partial ordering for which every pair of elements are comparable

totally (or linearly) ordered set: a poset with a total (or linear) ordering

well-ordered set: a poset $(S, \preccurlyeq)$, where $\preccurlyeq$ is a total order and every nonempty subset of S has a least element

lexicographic order: a partial ordering of Cartesian products or strings

Hasse diagram: a graphical representation of a poset where loops and all edges resulting from the transitive property are not shown, and the direction of the edges is indicated by the position of the vertices

maximal element: an element of a poset that is not less than any other element of the poset

minimal element: an element of a poset that is not greater than any other element of the poset

greatest element: an element of a poset greater than all other elements in this set

least element: an element of a poset less than all other elements in this set

upper bound of a set: an element in a poset greater than all other elements in the set

lower bound of a set: an element in a poset less than all other elements in the set

least upper bound of a set: an upper bound of the set that is less than all other upper bounds

greatest lower bound of a set: a lower bound of the set that is greater than all other lower bounds

lattice: a partially ordered set in which every two elements have a greatest lower bound and a least upper bound

compatible total ordering for a partial ordering: a total ordering that contains the given partial ordering

topological sort: the construction of a total ordering compatible with a given partial ordering

RESULTS

The reflexive closure of a relation R on the set A equals $R \cup \Delta$, where $\Delta = \{(a, a) \mid a \in A\}$.

The symmetric closure of a relation R on the set A equals $R \cup R^{-1}$, where $R^{-1} = \{(b, a) \mid (a, b) \in R\}$.

The transitive closure of a relation equals the connectivity relation formed from this relation.

Warshall's algorithm for finding the transitive closure of a relation

Let R be an equivalence relation. Then the following three statements are equivalent: (1) $a \, R \, b$; (2) $[a]_R \cap [b]_R \neq \emptyset$; (3) $[a]_R = [b]_R$.

The equivalence classes of an equivalence relation on a set A form a partition of A. Conversely, an equivalence relation can be constructed from any partition so that the equivalence classes are the subsets in the partition.

The principle of well-ordered induction

The topological sorting algorithm

Graphs

Chapter 10

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Chapter Summary

10.1 Graphs and Graph Models

10.2 Graph Terminology and Special Types of Graphs

10.3 Representing Graphs and Graph Isomorphism

10.4 Connectivity

10.5 Euler and Hamilton Paths

10.6 Shortest-Path Problems

10.7 Planar Graphs

10.8 Graph Coloring

10.2.2 Basic Terminology

To keep track of how many edges are incident to a vertex:

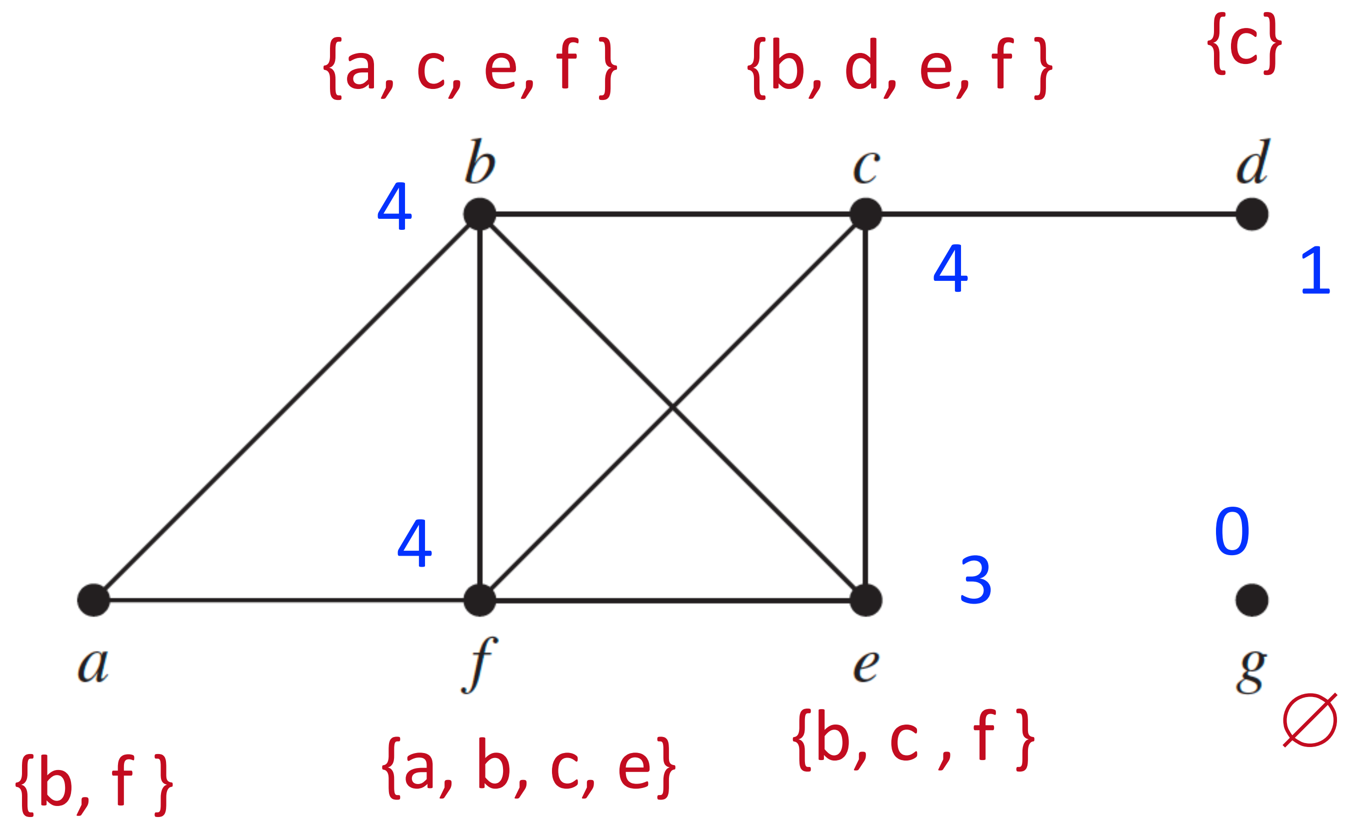
DEFINITION 3: The *degree of a vertex in a undirected graph* is the number of edges incident with it, except that a loop at a vertex contributes two to the degree of that vertex. The degree of the vertex v is denoted by $\deg(v)$.

Degrees and Neighborhoods of Vertices

Example: What are the degrees and neighborhoods of the vertices in the graphs?

deg

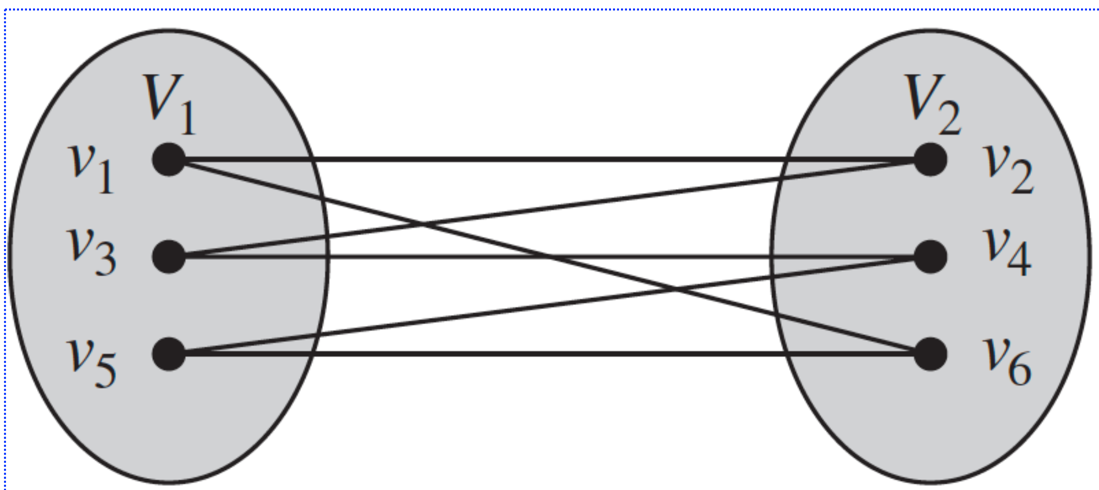
N(A)



10.2.4 Bipartite Graphs

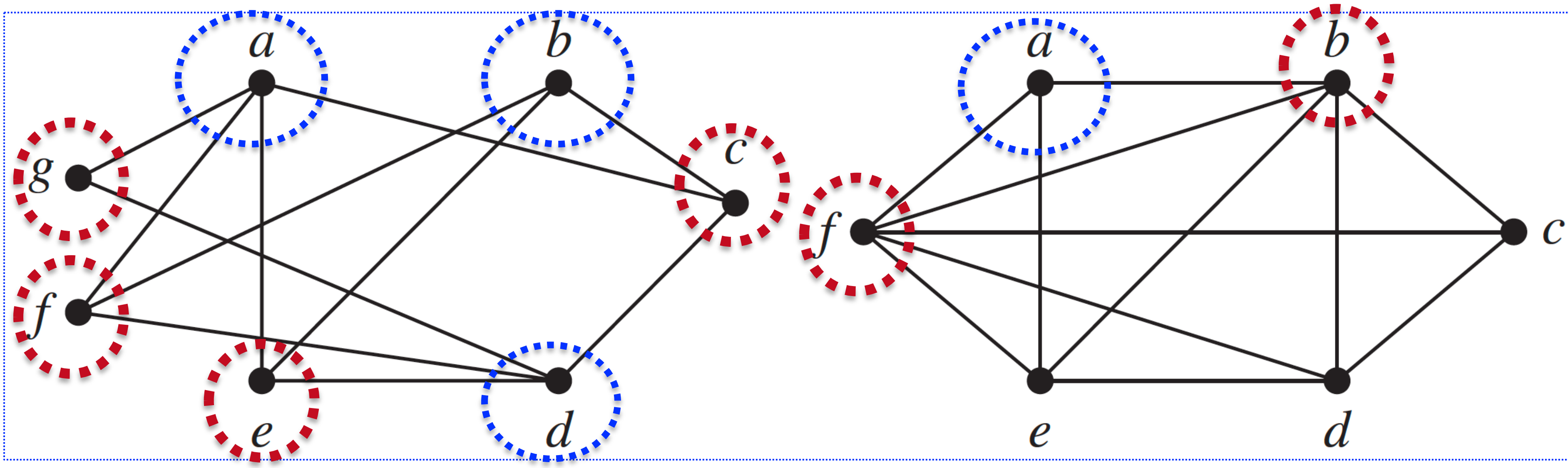
Sometimes a graph has the property that its vertex set can be divided into two disjoint subsets such that each edge connects a vertex in one of these subsets to a vertex in the other subset.

DEFINITION 6: A simple graph G is called bipartite if its vertex set V can be partitioned into two disjoint sets V_1 and V_2 such that every edge in the graph connects a vertex in V_1 and a vertex in V_2 (so that no edge in G connects either two vertices in V_1 or two vertices in V_2). When this condition holds, we call the pair (V_1, V_2) a bipartition of the vertex set V of G.



Bipartite Graphs

It is a bipartite. This is not bipartite.

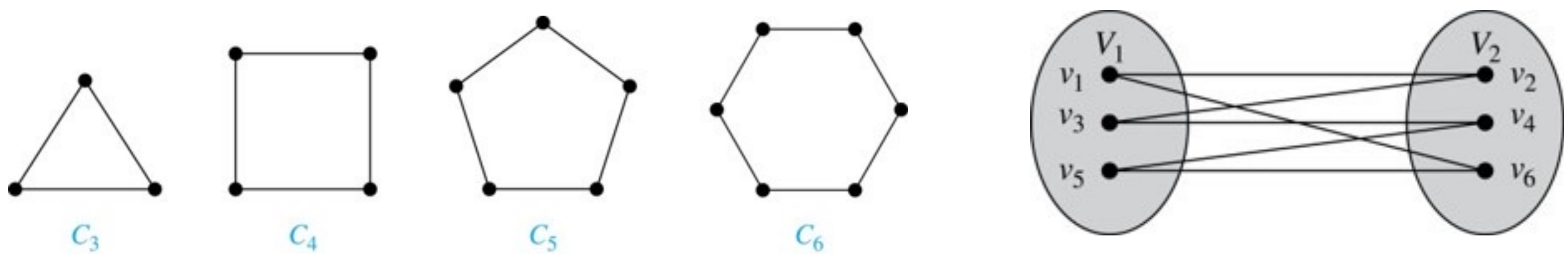


Theorem 4: A simple graph is bipartite if and only if it is possible to assign one of two different colors to each vertex of the graph so that no two adjacent vertices are assigned the same color.

Bipartite Graphs

Example: Show that C_6 is bipartite.

Solution: We can partition the vertex set into $V_1 = \{v_1, v_3, v_5\}$ and $V_2 = \{v_2, v_4, v_6\}$ so that every edge of C_6 connects a vertex in V_1 and V_2 .

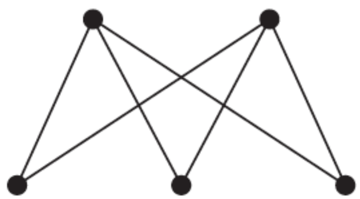


Example: Show that C_3 is not bipartite.

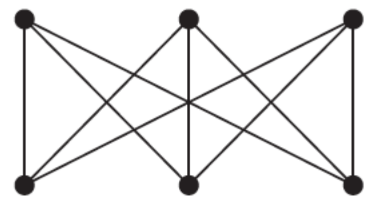
Solution: If we divide the vertex set of C_3 into two nonempty sets, one of the two must contain two vertices. But in C_3 every vertex is connected to every other vertex. Therefore, the two vertices in the same partition are connected. Hence, C_3 is not bipartite.

Complete Bipartite Graphs

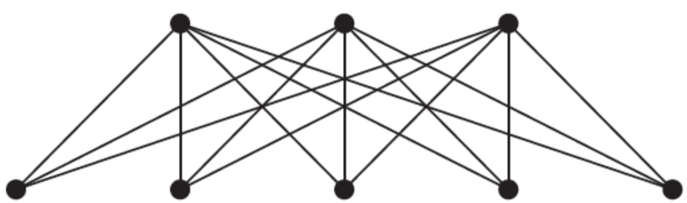
A complete bipartite graph $K_{m,n}$ is a graph that has its vertex set partitioned into two subsets of m and n vertices, respectively with an edge between two vertices if and only if one vertex is in the first subset and the other vertex is in the second subset.



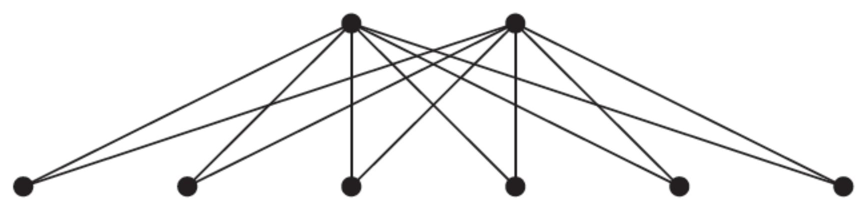
$K_{2,3}$



$K_{3,3}$



$K_{3,5}$



$K_{2,6}$

Isomorphism and Nonisomorphic

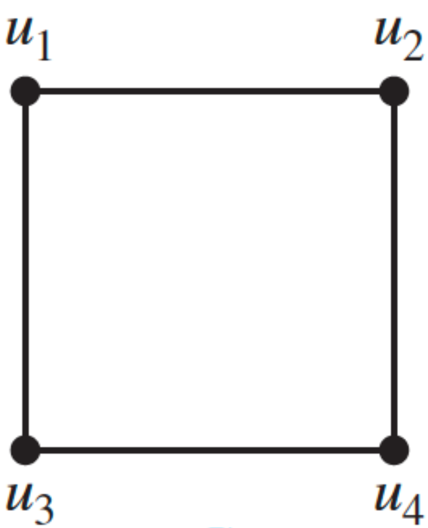
DEFINITION 1:The simple graphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ are isomorphic if there exists a one-to one and onto function f from V_1 to V_2 with the property that a and b are adjacent in G_1 if and only if f (a) and f (b) are adjacent in G_2, for all a and b in V_1. Such a function f is called an isomorphism. Two simple graphs that are not isomorphic are called nonisomorphic.

In other words, when two simple graphs are isomorphic, there is a one-to-one correspondence between vertices of the two graphs that preserves the adjacency relationship.

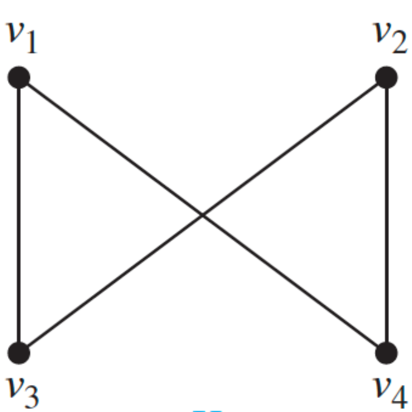
Isomorphism of simple graphs is an equivalence relation.

Isomorphism

Example: Show that the graphs G = (V, E) and H = (W, F) are isomorphic.



G



H

Solution:

The function f with $f(u_1) = v_1$, $f(u_2) = v_4$, $f(u_3) = v_3$, and $f(u_4) = v_2$ is a one-to-one correspondence between V and W.

Note that adjacent vertices in G are u_1 and u_2, u_1 and u_3, u_2 and u_4, and u_3 and u_4.

Each of the pairs $f(u_1) = v_1$ and $f(u_2) = v_4$, $f(u_1) = v_1$ and $f(u_3) = v_3$, $f(u_2) = v_4$ and $f(u_4) = v_2$, and $f(u_3) = v_3$ and $f(u_4) = v_2$ consists of two adjacent vertices in H.

10.3.6 Determining whether Two Simple Graphs are Isomorphic

It is often difficult to determine whether two simple graphs are isomorphic.

- There are $n!$ possible one-to-one correspondences between the vertex sets of two simple graphs with n vertices.
- Testing each such correspondence to see whether it preserves adjacency and nonadjacency is impractical if n is at all large.

Sometimes it is not hard to show that two graphs are not isomorphic.

- In particular, we can show that two graphs are not isomorphic if we can find a property only one of the two graphs has, but that is preserved by isomorphism.

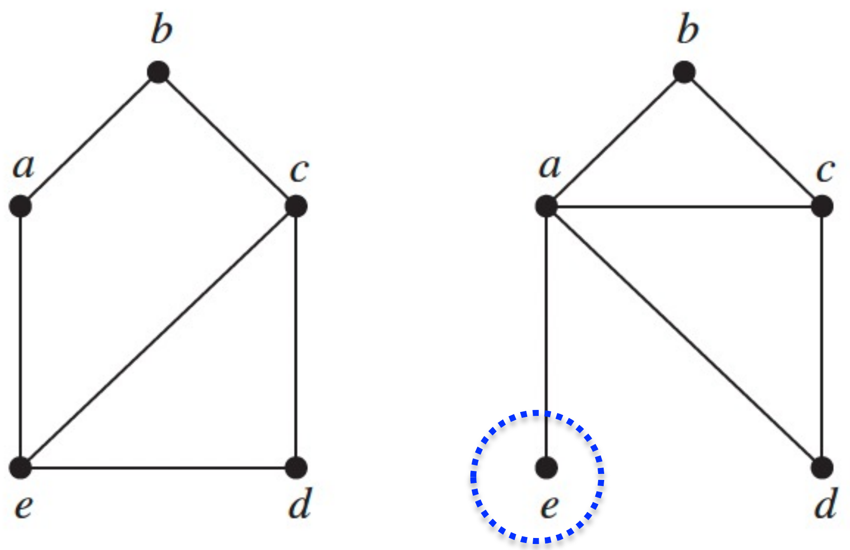
10.3.6 Determining whether Two Simple Graphs are Isomorphic

A property preserved by isomorphism of graphs is called a graph invariant.

- For instance, isomorphic simple graphs must have the same number of vertices, because there is a one-to-one correspondence between the sets of vertices of the graphs.
- Isomorphic simple graphs also must have the same number of edges, because the one-to one correspondence between vertices establishes a one-to-one correspondence between edges.
- In addition, the degrees of the vertices in isomorphic simple graphs must be the same.

Determining whether Two Simple Graphs are Isomorphic

Example: Show that the graphs are not isomorphic.



G H

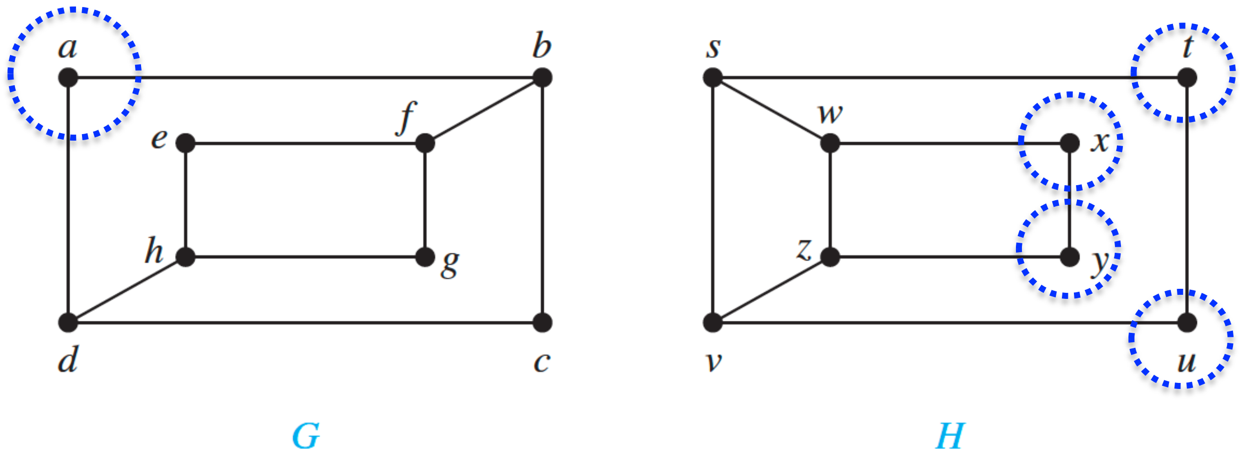
Both G and H have five vertices and six edges.

H has a vertex (e) of degree one, whereas G has no vertices of degree one.

It follows that G and H are not isomorphic.

Determining whether Two Simple Graphs are Isomorphic

Example: Determine whether the graphs are isomorphic?

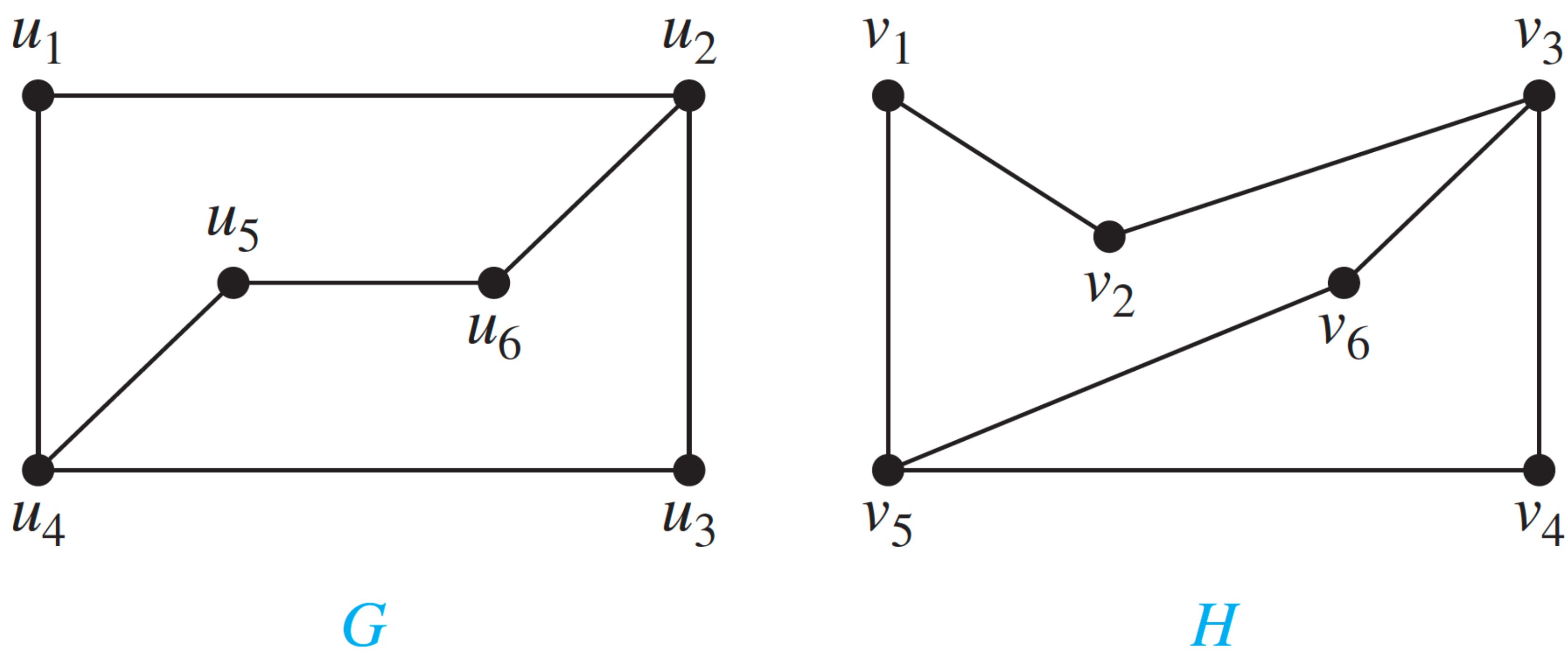


The graphs G and H both have eight vertices and 10 edges. They also both have four vertices of degree two and four of degree three.

However, G and H are not isomorphic. Note that since deg(a) = 2 in G, a must correspond to t, u, x, or y in H, because these are the vertices of degree 2. But each of these vertices is adjacent to another vertex of degree two in H, which is not true for a in G.

Determining whether Two Simple Graphs are Isomorphic

Example: Determine whether the graphs G and H are isomorphic?



Determining whether Two Simple Graphs are Isomorphic

Example: Determine whether the graphs G and H are isomorphic?

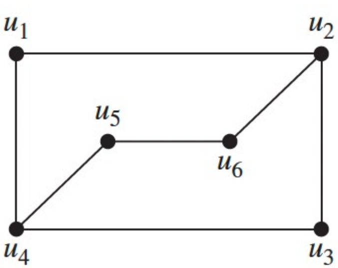
Solution: Both graphs have six vertices and seven edges.

They also both have four vertices of degree two and two of degree three.

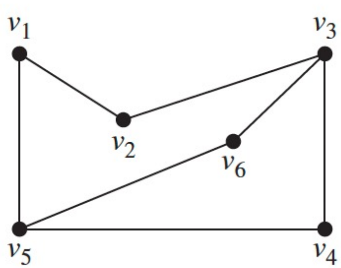
The subgraphs of *G* and *H* consisting of all the vertices of degree two and the edges connecting them are isomorphic. So, it is reasonable to try to find an isomorphism *f*.

The function *f* with $f(u_1) = v_6$, $f(u_2) = v_3$, $f(u_3) = v_4$, and $f(u_4) = v_5$, $f(u_5) = v_1$, and $f(u_6) = v_2$ is a one-to-one correspondence between *G* and *H*.

Because *f* is an isomorphism, it follows that *G* and *H* are isomorphic graphs.



G



H

$$\mathbf{A}_G = \begin{array}{cc} & \begin{array}{cccccc} u_1 & u_2 & u_3 & u_4 & u_5 & u_6 \end{array} \\ \begin{array}{c} u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \\ u_6 \end{array} & \begin{bmatrix} 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 \end{bmatrix} \end{array}, \quad \mathbf{A}_H = \begin{array}{cc} & \begin{array}{cccccc} v_6 & v_3 & v_4 & v_5 & v_1 & v_2 \end{array} \\ \begin{array}{c} v_6 \\ v_3 \\ v_4 \\ v_5 \\ v_1 \\ v_2 \end{array} & \begin{bmatrix} 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 \end{bmatrix} \end{array}.$$

10.4.6 Paths and Isomorphism

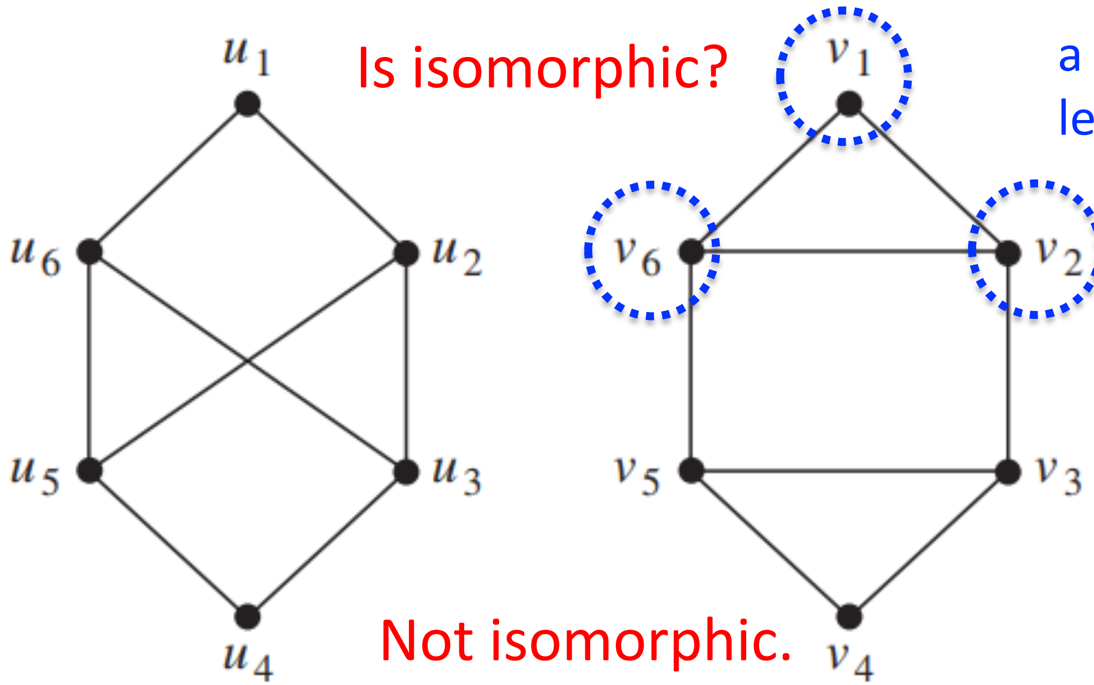
There are several ways that <u>paths and circuits can help</u> determine whether two graphs are isomorphic.

- ✓ The existence of a simple circuit of a particular length is a useful invariant that can be used to show that two graphs are not isomorphic.
- ✓ The paths can be used to construct mappings that may be isomorphisms.

six vertices

eight edges

Is isomorphic?



a simple circuit of length three

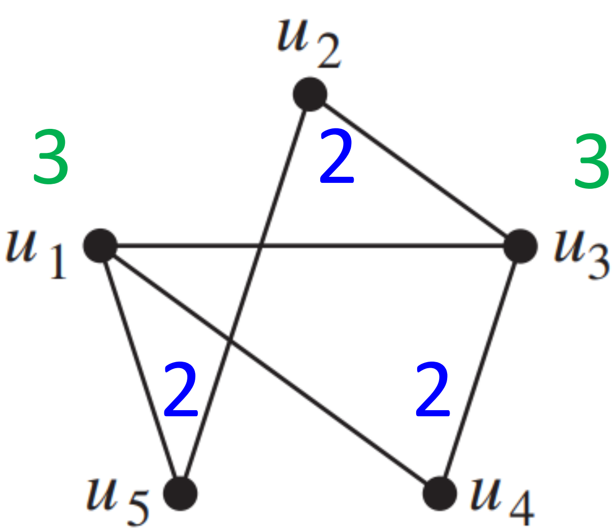
Not isomorphic.

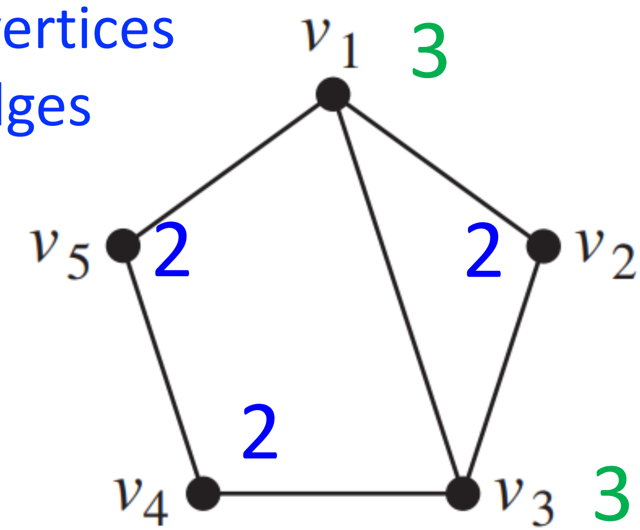
Paths and Isomorphism

Example: Determine whether the graphs are isomorphic.

Five vertices

Six edges





Degree two : three vertices

Degree three: two vertices

A simple circuit of length three

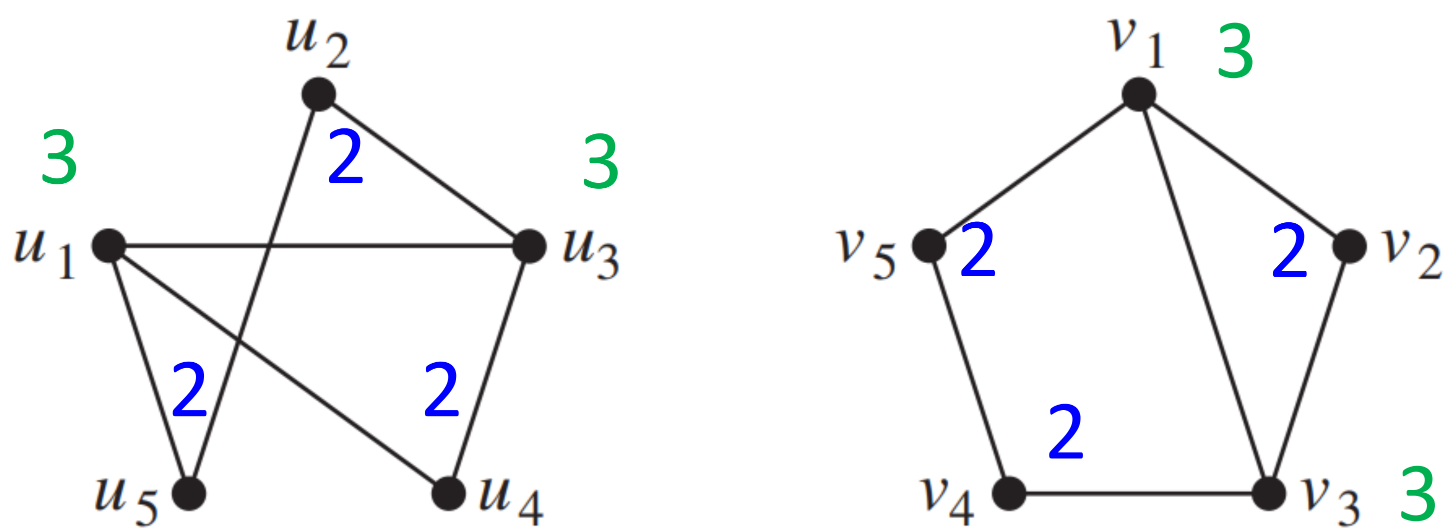
A simple circuit of length four

A simple circuit of length five

Because all these isomorphic invariants agree, so they may be isomorphic.

Paths and Isomorphism

Example: Determine whether the graphs are isomorphic.



To find a possible isomorphism, we can follow paths that go through all vertices so that the corresponding vertices in the two graphs have the same degree.

start at a vertex of degree three; go through vertices of degrees two, three, and two, respectively; and end at a vertex of degree two.

By following these paths through the graphs, we define the mapping f with f (u1) = v3, f (u4) = v2, f (u3) = v1, f (u2) = v5, and f (u5) = v4.

Euler and Hamiltonian Graphs

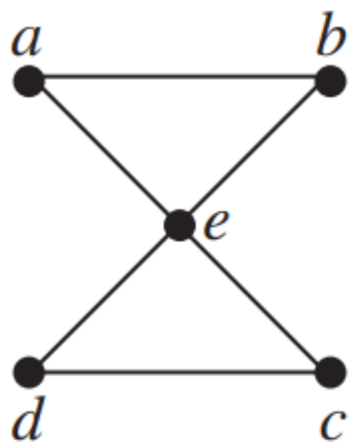
Section 10.5

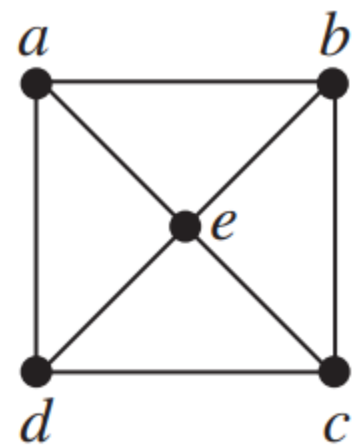
Euler Paths and Circuits

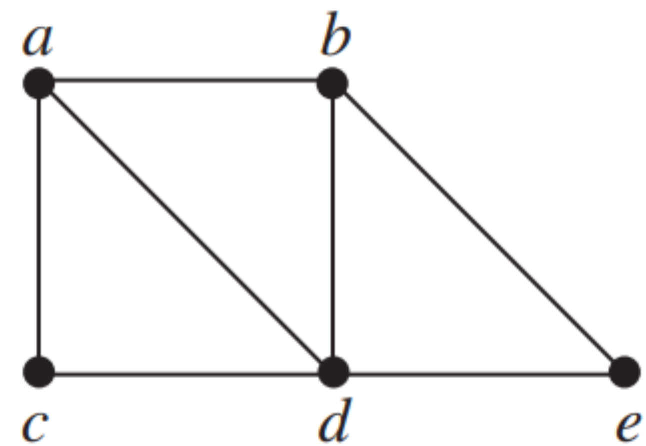
DEFINITION 1: An Euler circuit in a graph G is a simple circuit containing every edge of G. An Euler path in G is a simple path containing every edge of G.

Euler circuit

Yes No No







Yes No Yes

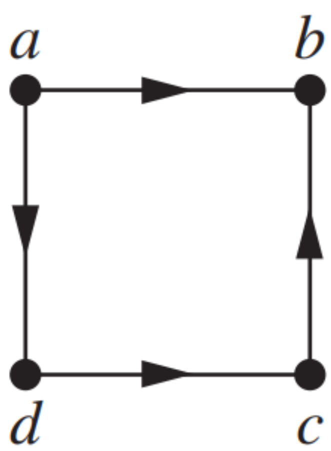
Euler path

Euler Paths and Circuits

Example: Which of the directed graphs have an Euler circuit? Of those that do not, which have an Euler path?

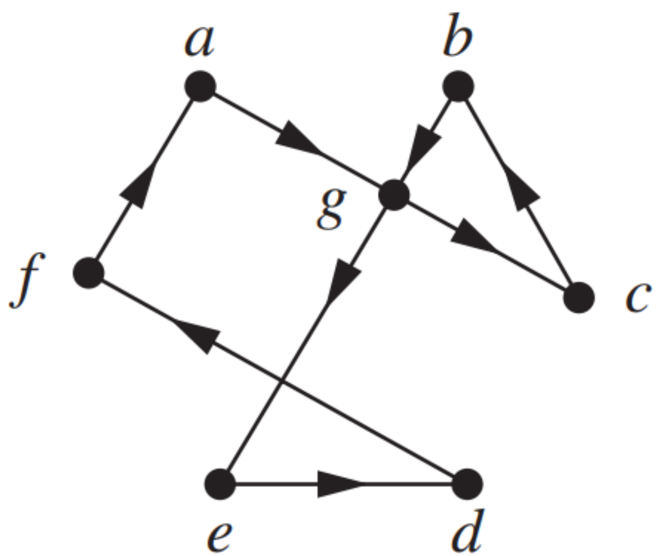
Euler circuit

No

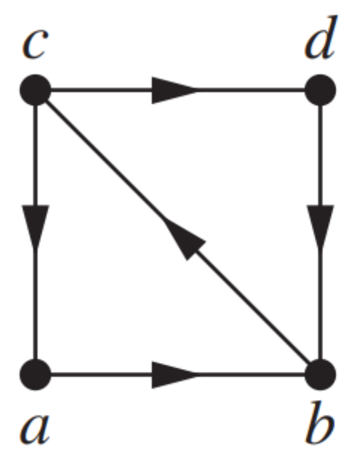


No

Yes



No

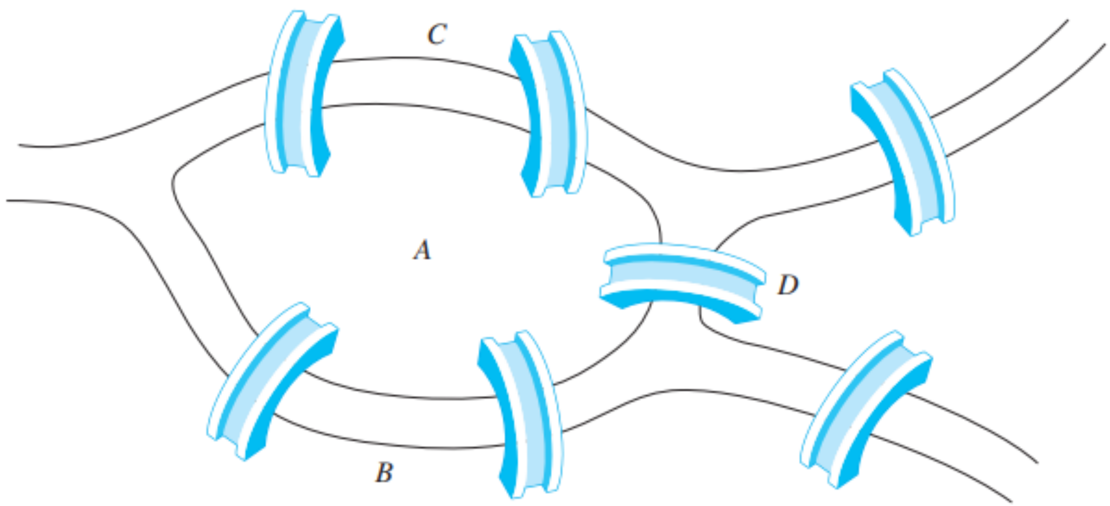


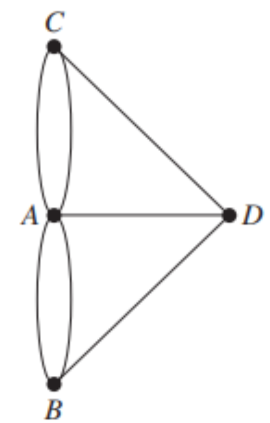
Yes

Euler path

Necessary and Sufficient Conditions for Euler Circuits

THEOREM 1: A connected multigraph with at least two vertices has an Euler circuit if and only if each of its vertices has even degree.



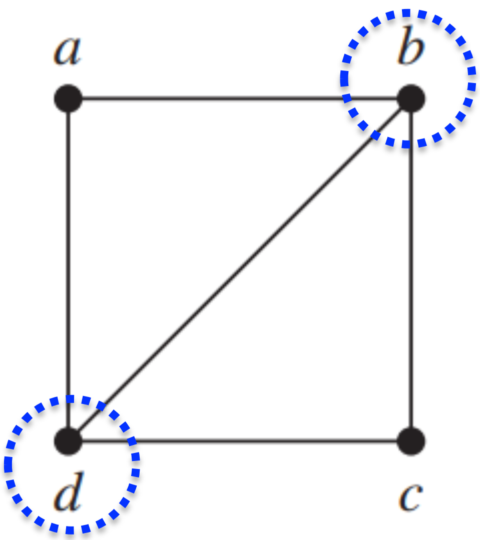


The graph has four vertices of odd degree, it does not have an Euler circuit.

There is no way to start at a given point, cross each bridge exactly once, and return to the starting point.

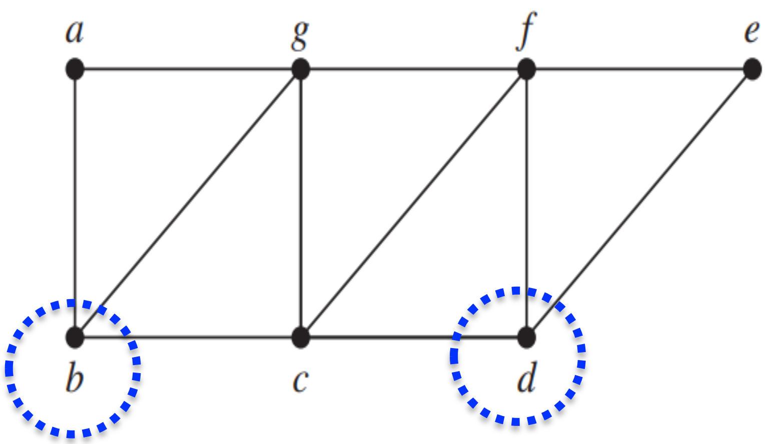
Necessary and Sufficient Conditions for Euler Paths

THEOREM 2: A connected multigraph has an Euler path but not an Euler circuit if and only if it has exactly two vertices of odd degree.



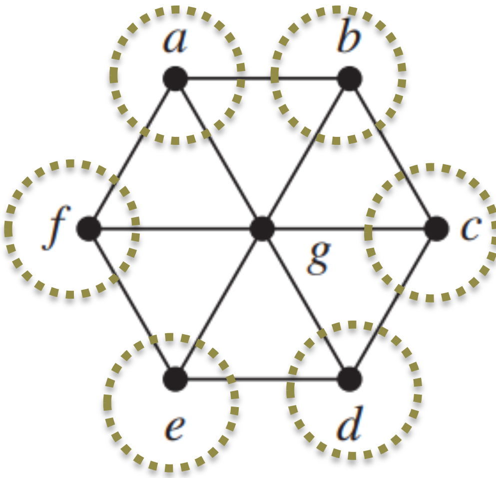
has an Euler path

e.g., *d*, *a*, *b*, *c*, *d*, *b*.



has an Euler path

e.g., *b*, *a*, *g*, *f*, *e*, *d*, *c*, *g*, *b*, *c*, *f*, *d*.



does not have an Euler path

Key Terms and Results

TERMS

undirected edge: an edge associated to a set $\{u, v\}$, where u and v are vertices

directed edge: an edge associated to an ordered pair (u, v), where u and v are vertices

multiple edges: distinct edges connecting the same vertices

multiple directed edges: distinct directed edges associated with the same ordered pair (u,v), where u and v are vertices

loop: an edge connecting a vertex with itself

undirected graph: a set of vertices and a set of undirected edges each of which is associated with a set of one or two of these vertices

simple graph: an undirected graph with no multiple edges or loops

multigraph: an undirected graph that may contain multiple edges but no loops

pseudograph: an undirected graph that may contain multiple edges and loops

directed graph: a set of vertices together with a set of directed edges each of which is associated with an ordered pair of vertices

directed multigraph: a graph with directed edges that may contain multiple directed edges

simple directed graph: a directed graph without loops or multiple directed edges

adjacent: two vertices are adjacent if there is an edge between them

incident: an edge is incident with a vertex if the vertex is an endpoint of that edge

deg v (degree of the vertex v in an undirected graph): the number of edges incident with v with loops counted twice

$\deg^-(v)$ (the in-degree of the vertex v in a graph with directed edges): the number of edges with v as their terminal vertex

$\deg^+(v)$ (the out-degree of the vertex v in a graph with directed edges): the number of edges with v as their initial vertex

underlying undirected graph of a graph with directed edges: the undirected graph obtained by ignoring the directions of the edges

K_n (complete graph on n vertices): the undirected graph with n vertices where each pair of vertices is connected by an edge

bipartite graph: a graph with vertex set that can be partitioned into subsets V_1 and V_2 so that each edge connects a vertex in V_1 and a vertex in V_2. The pair (V_1, V_2) is called a **bipartition** of V.

$K_{m,n}$ (complete bipartite graph): the graph with vertex set partitioned into a subset of m elements and a subset of n elements with two vertices connected by an edge if and only if one is in the first subset and the other is in the second subset

Key Terms and Results

C_n (cycle of size n), $n \geq 3$: the graph with n vertices $v_1, v_2, \ldots, v_n$ and edges $\{v_1, v_2\}$, $\{v_2, v_3\}$, …, $\{v_{n-1}, v_n\}$, $\{v_n, v_1\}$

W_n (wheel of size n), $n \geq 3$: the graph obtained from C_n by adding a vertex and edges from this vertex to the original vertices in C_n

Q_n (n-cube), $n \geq 1$: the graph that has the 2^n bit strings of length n as its vertices and edges connecting every pair of bit strings that differ by exactly one bit

matching in a graph G: a set of edges such that no two edges have a common endpoint

complete matching M from V_1 to V_2: a matching such that every vertex in V_1 is an endpoint of an edge in M

maximum matching: a matching containing the most edges among all matchings in a graph

isolated vertex: a vertex of degree zero

pendant vertex: a vertex of degree one

regular graph: a graph where all vertices have the same degree

subgraph of a graph $G = (V, E)$: a graph (W, F), where W is a subset of V and F is a subset of E

$G_1 \cup G_2$ (union of G_1 and G_2): the graph $(V_1 \cup V_2, E_1 \cup E_2)$, where $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$

adjacency matrix: a matrix representing a graph using the adjacency of vertices

incidence matrix: a matrix representing a graph using the incidence of edges and vertices

isomorphic simple graphs: the simple graphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ are isomorphic if there exists a one-to-one correspondence f from V_1 to V_2 such that $\{f(v_1), f(v_2)\} \in E_2$ if and only if $\{v_1, v_2\} \in E_1$ for all v_1 and v_2 in V_1

invariant for graph isomorphism: a property that isomorphic graphs either both have or both do not have

path from u to v in an undirected graph: a sequence of edges $e_1, e_2, \ldots, e_n$, where e_i is associated to $\{x_i, x_{i+1}\}$ for $i = 0, 1, \ldots, n$, where $x_0 = u$ and $x_{n+1} = v$

path from u to v in a graph with directed edges: a sequence of edges $e_1, e_2, \ldots, e_n$, where e_i is associated to (x_i, x_{i+1}) for $i = 0, 1, \ldots, n$, where $x_0 = u$ and $x_{n+1} = v$

simple path: a path that does not contain an edge more than once

circuit: a path of length $n \geq 1$ that begins and ends at the same vertex

connected graph: an undirected graph with the property that there is a path between every pair of vertices

cut vertex of G: a vertex v such that $G - v$ is disconnected

cut edge of G: an edge e such that $G - e$ is disconnected

nonseparable graph: a graph without a cut vertex

vertex cut of G: a subset V' of the set of vertices of G such that $G - V'$ is disconnected

$\kappa(G)$ (the vertex connectivity of G): the size of a smallest vertex cut of G

Key Terms and Results

k-connected graph: a graph that has a vertex connectivity no smaller than k

edge cut of G: a set of edges E' of G such that $G - E'$ is disconnected

$\lambda(G)$ (the edge connectivity of G): the size of a smallest edge cut of G

connected component of a graph G: a maximal connected subgraph of G

strongly connected directed graph: a directed graph with the property that there is a directed path from every vertex to every vertex

strongly connected component of a directed graph G: a maximal strongly connected subgraph of G

Euler path: a path that contains every edge of a graph exactly once

Euler circuit: a circuit that contains every edge of a graph exactly once

Hamilton path: a path in a graph that passes through each vertex exactly once

Hamilton circuit: a circuit in a graph that passes through each vertex exactly once

weighted graph: a graph with numbers assigned to its edges

shortest-path problem: the problem of determining the path in a weighted graph such that the sum of the weights of the edges in this path is a minimum over all paths between specified vertices

traveling salesperson problem: the problem that asks for the circuit of shortest total length that visits every vertex of a weghted graph exactly once

planar graph: a graph that can be drawn in the plane with no crossings

regions of a representation of a planar graph: the regions the plane is divided into by the planar representation of the graph

elementary subdivision: the removal of an edge $\{u, v\}$ of an undirected graph and the addition of a new vertex w together with edges $\{u, w\}$ and $\{w, v\}$

homeomorphic: two undirected graphs are homeomorphic if they can be obtained from the same graph by a sequence of elementary subdivisions

graph coloring: an assignment of colors to the vertices of a graph so that no two adjacent vertices have the same color

chromatic number: the minimum number of colors needed in a coloring of a graph

RESULTS

The handshaking theorem: If $G = (V, E)$ be an undirected graph with m edges, then $2m = \sum_{v \in V} \deg(v)$.

Hall's marriage theorem: The bipartite graph $G = (V, E)$ with bipartition (V_1, V_2) has a complete matching from V_1 to V_2 if and only if $|N(A)| \geq |A|$ for all subsets A of V_1.

There is an Euler circuit in a connected multigraph if and only if every vertex has even degree.

There is an Euler path in a connected multigraph if and only if at most two vertices have odd degree.

Dijkstra's algorithm: a procedure for finding a shortest path between two vertices in a weighted graph (see Section 10.6)

Euler's formula: $r = e - v + 2$ where r, e, and v are the number of regions of a planar representation, the number of edges, and the number of vertices, respectively, of a connected planar graph

Kuratowski's theorem: A graph is nonplanar if and only if it contains a subgraph homeomorphic to $K_{3,3}$ or K_5. (Proof beyond the scope of this book.)

The four color theorem: Every planar graph can be colored using no more than four colors. (Proof far beyond the scope of this book!)

Boolean Algebra

Chapter 12

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Boolean algebra

Boolean algebra	logical operators
complementation($\bar{\ }$)	$\neg$
Boolean sum(+)	$\vee$
Boolean product($\cdot$)	$\wedge$
0	**F** (false)
1	**T** (true)

$$1 \cdot 0 + \overline{(0 + 1)} = 0, \quad \Longleftrightarrow \quad (\mathbf{T} \wedge \mathbf{F}) \vee \neg(\mathbf{T} \vee \mathbf{F}) \equiv \mathbf{F}.$$

12.1.3 Identities of Boolean Algebra

TABLE 5 Boolean Identities.

Identity	*Name*
$\overline{\overline{x}} = x$	Law of the double complement
$x + x = x$ $x \cdot x = x$	Idempotent laws
$x + 0 = x$ $x \cdot 1 = x$	Identity laws
$x + 1 = 1$ $x \cdot 0 = 0$	Domination laws
$x + y = y + x$ $xy = yx$	Commutative laws
$x + y(y + z) = (x + y) + z$ $x(yz) = (xy)z$	Associative laws
$x + yz = (x + y) + (x + z)$ $x(y + z) = xy + xz$	Distributive laws
$\overline{(xy)} = \overline{x} + \overline{y}$ $\overline{(x + y)} = \overline{x}\,\overline{y}$	De Morgan's laws
$x + xy = x$ $x(x + y) = x$	Absorption laws
$x + \overline{x} = 1$	Unit property
$x\overline{x} = 0$	Zero property

Each identity can be proved using a table.

All identities in Table 5, except for the first and the last two come in pairs. Each element of the pair is the *dual* of the other (obtained by switching Boolean sums and Boolean products and 0 and 1.

The Boolean identities correspond to the identities of propositional logic (Section 1.3) and the set identities (Section 2.2).

Key Terms and Results

TERMS

Boolean variable: a variable that assumes only the values 0 and 1

$\overline{x}$ (complement of x): an expression with the value 1 when x has the value 0 and the value 0 when x has the value 1

$x \cdot y$ (or xy) (Boolean product or conjunction of x and y): an expression with the value 1 when both x and y have the value 1 and the value 0 otherwise

$x + y$ (Boolean sum or disjunction of x and y): an expression with the value 1 when either x or y, or both, has the value 1, and 0 otherwise

Boolean expressions: the expressions obtained recursively by specifying that $0, 1, x_1, \ldots, x_n$ are Boolean expressions and $\overline{E}_1$, $(E_1 + E_2)$, and (E_1E_2) are Boolean expressions if E_1 and E_2 are

dual of a Boolean expression: the expression obtained by interchanging + signs and · signs and interchanging 0s and 1s

Boolean function of degree n: a function from B^n to B where $B = \{0, 1\}$

Boolean algebra: a set B with two binary operations $\vee$ and $\wedge$, elements 0 and 1, and a complementation operator $^-$ that satisfies the identity, complement, associative, commutative, and distributive laws

literal of the Boolean variable x: either x or $\overline{x}$

minterm of $x_1, x_2, \ldots, x_n$: a Boolean product $y_1y_2 \cdots y_n$, where each y_i is either x_i or $\overline{x}_i$

sum-of-products expansion (or disjunctive normal form): the representation of a Boolean function as a disjunction of minterms

functionally complete: a set of Boolean operators is called functionally complete if every Boolean function can be represented using these operators

$x \mid y$ (or x NAND y): the expression that has the value 0 when both x and y have the value 1 and the value 1 otherwise

$x \downarrow y$ (or x NOR y): the expression that has the value 0 when either x or y or both have the value 1 and the value 0 otherwise

inverter: a device that accepts the value of a Boolean variable as input and produces the complement of the input

OR gate: a device that accepts the values of two or more Boolean variables as input and produces their Boolean sum as output

AND gate: a device that accepts the values of two or more Boolean variables as input and produces their Boolean product as output

half adder: a circuit that adds two bits, producing a sum bit and a carry bit

full adder: a circuit that adds two bits and a carry, producing a sum bit and a carry bit

K-map for n variables: a rectangle divided into 2^n cells where each cell represents a minterm in the variables

minimization of a Boolean function: representing a Boolean function as the sum of the fewest products of literals such that these products contain the fewest literals possible among all sums of products that represent this Boolean function

implicant of a Boolean function: a product of literals with the property that if this product has the value 1, then the value of this Boolean function is 1

prime implicant of a Boolean function: a product of literals that is an implicant of the Boolean function and no product obtained by deleting a literal is also an implicant of this function

essential prime implicant of a Boolean function: a prime implicant of the Boolean function that must be included in a minimization of this function

don't care condition: a combination of input values for a circuit that is not possible or never occurs

Key Terms and Results

RESULTS

The identities for Boolean algebra (see Table 5 in Section 12.1).

An identity between Boolean functions represented by Boolean expressions remains valid when the duals of both sides of the identity are taken.

Every Boolean function can be represented by a sum-of-products expansion.

Each of the sets $\{+, {}^{-}\}$ and $\{\cdot, {}^{-}\}$ is functionally complete.

Each of the sets $\{\downarrow\}$ and $\{\mid\}$ is functionally complete.

The use of K-maps to minimize Boolean expressions.

The Quine–McCluskey method for minimizing Boolean expressions.